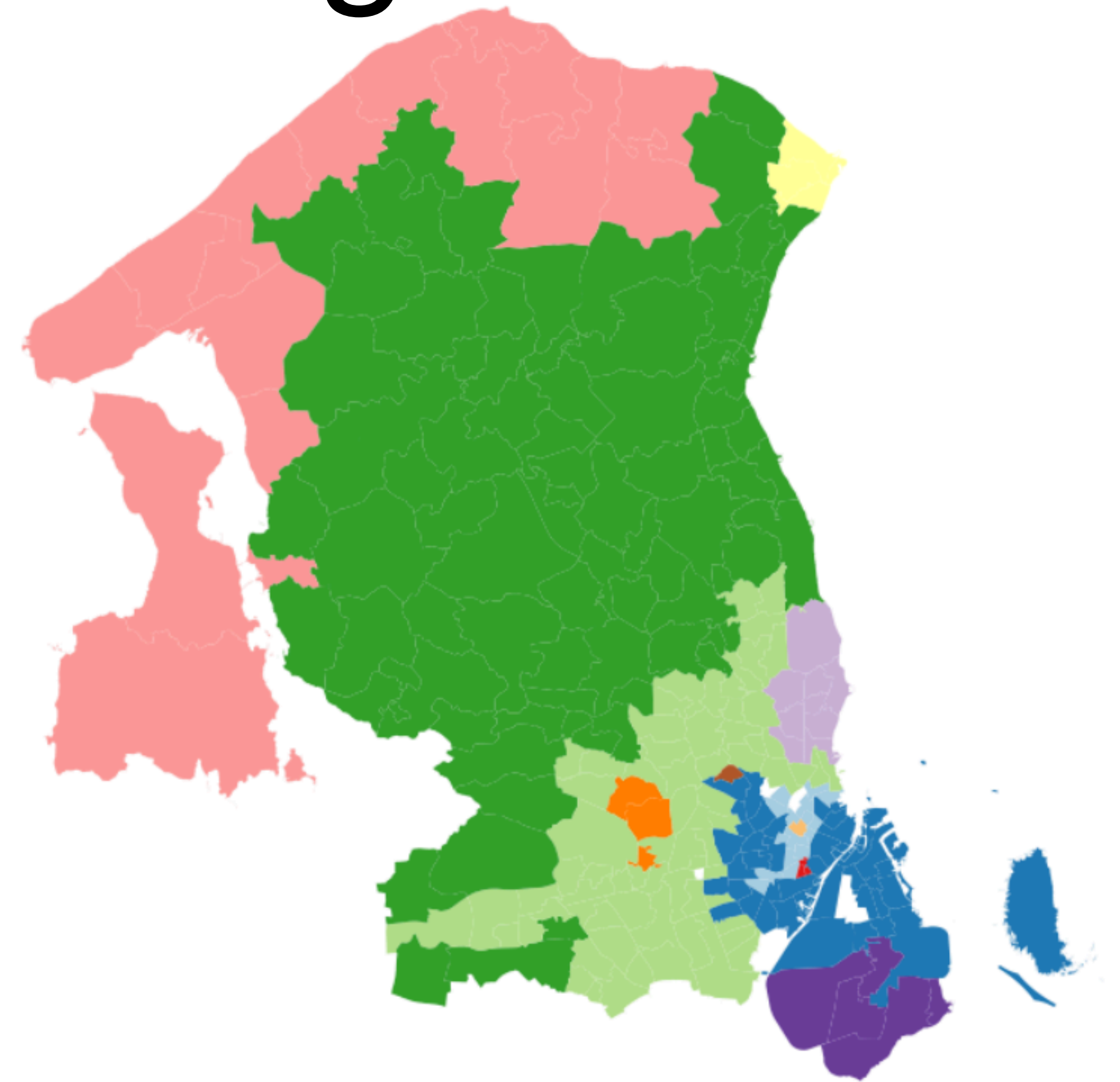


Lecture 7: Spatial clustering

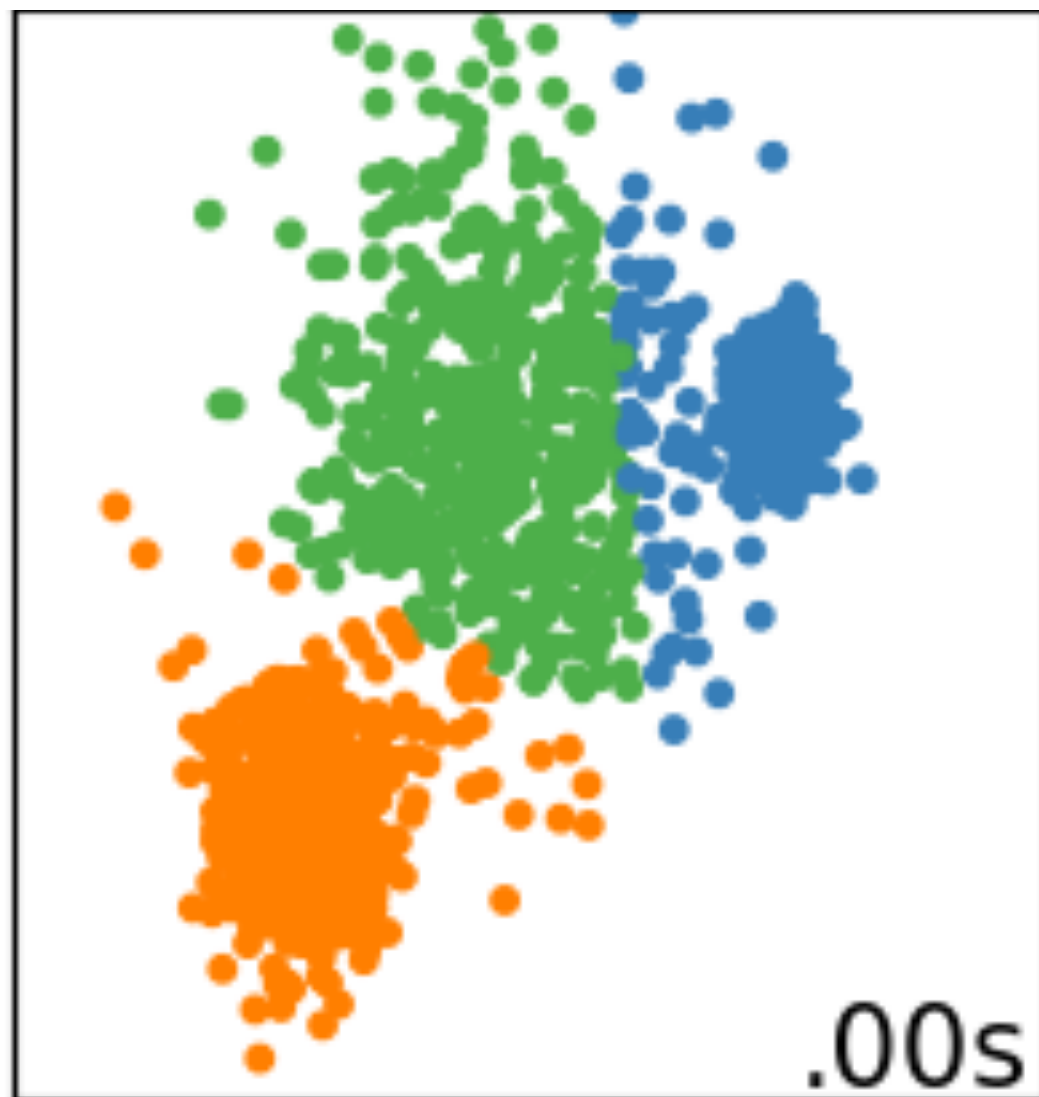
Instructor: Michael Szell

Mar 11, 2026



Today you will learn about spatial clustering

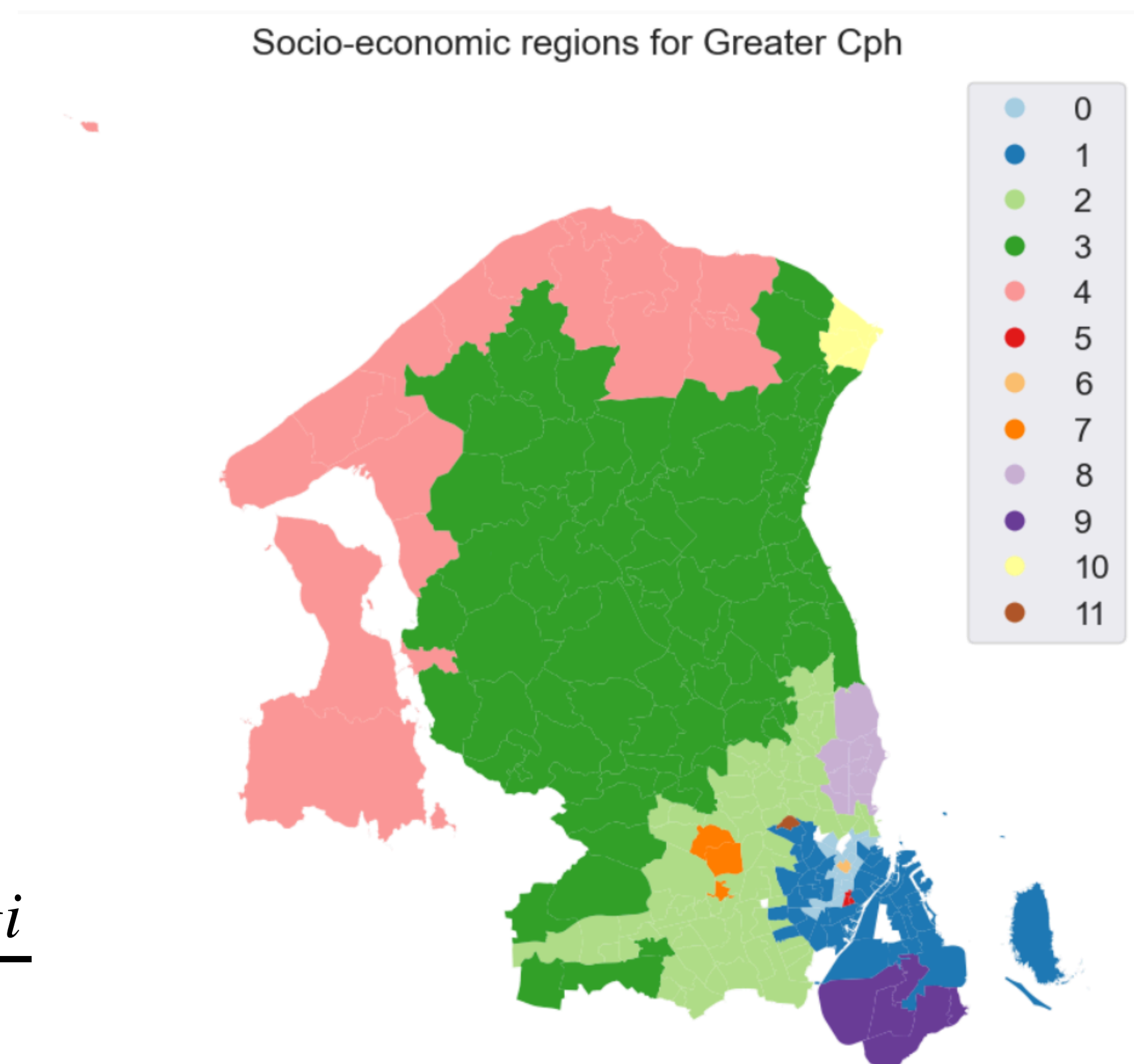
Clustering



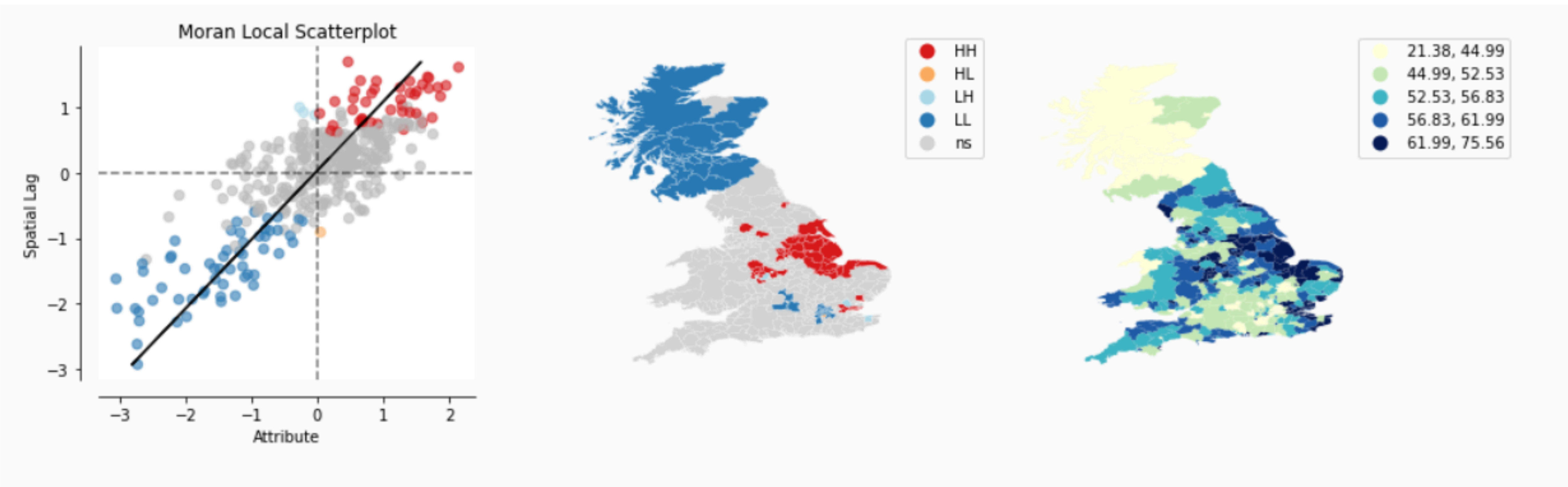
$$H = \frac{\sum_{i=1}^m u_i}{\sum_{i=1}^m w_i + \sum_{i=1}^m u_i}$$

$$IPQ_i = \frac{A_i}{A_c} = \frac{4\pi A_i}{P_i^2}$$

Regionalization with socioeconomic data



Last week: Spatial Autocorrelation



Cluster = Portion of a map where values are correlated in a particularly strong or specific way

The world is complex and multidimensional.

Univariate

Percent of foreign-born

Years of schooling

Monthly income

Multivariate

Neighborhood

Human development

Deprivation

Everything should be made as simple
as possible, but not simpler.

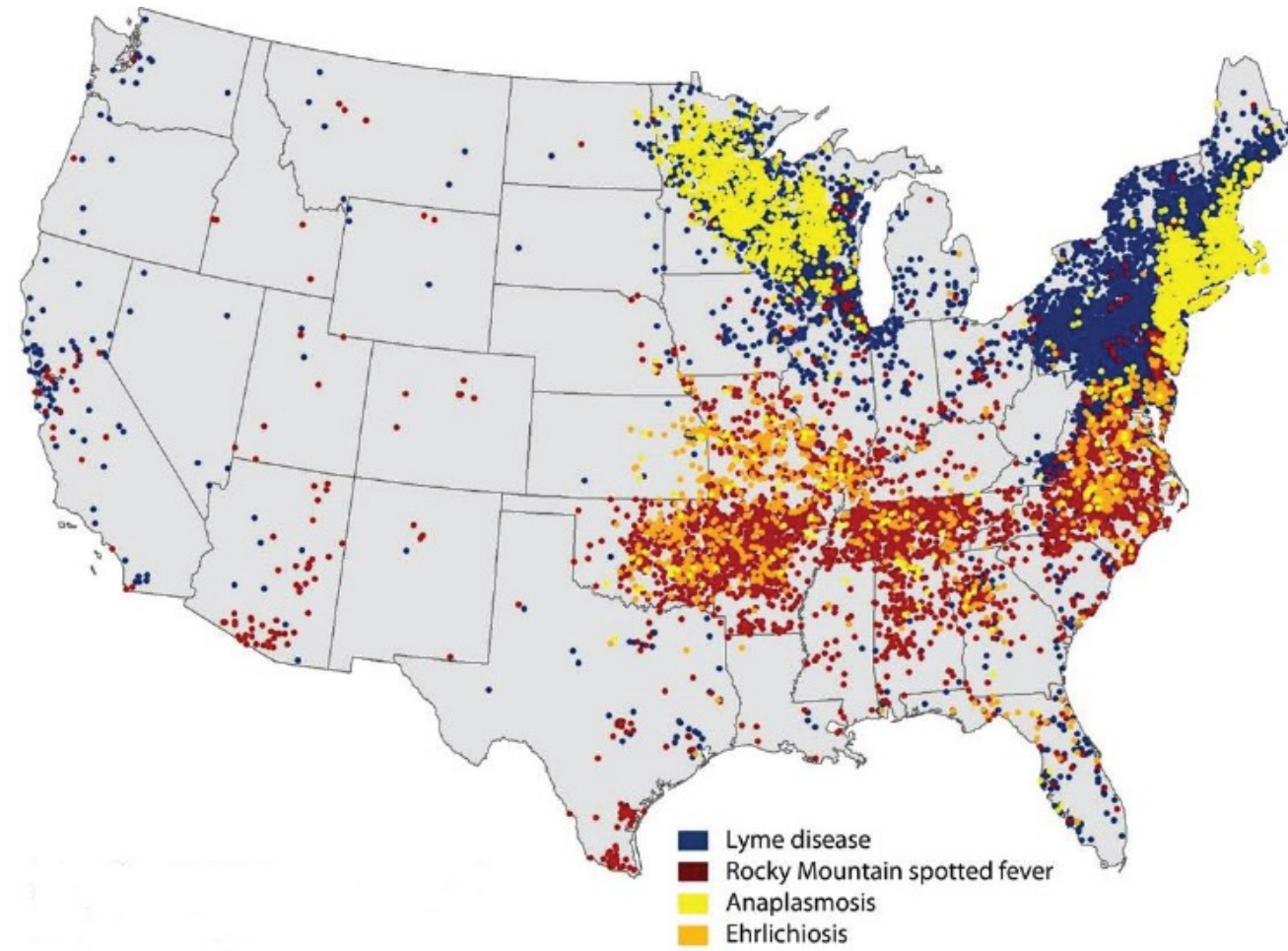
Albert Einstein

Cluster analysis is the division of data into groups that are meaningful, useful, or both.

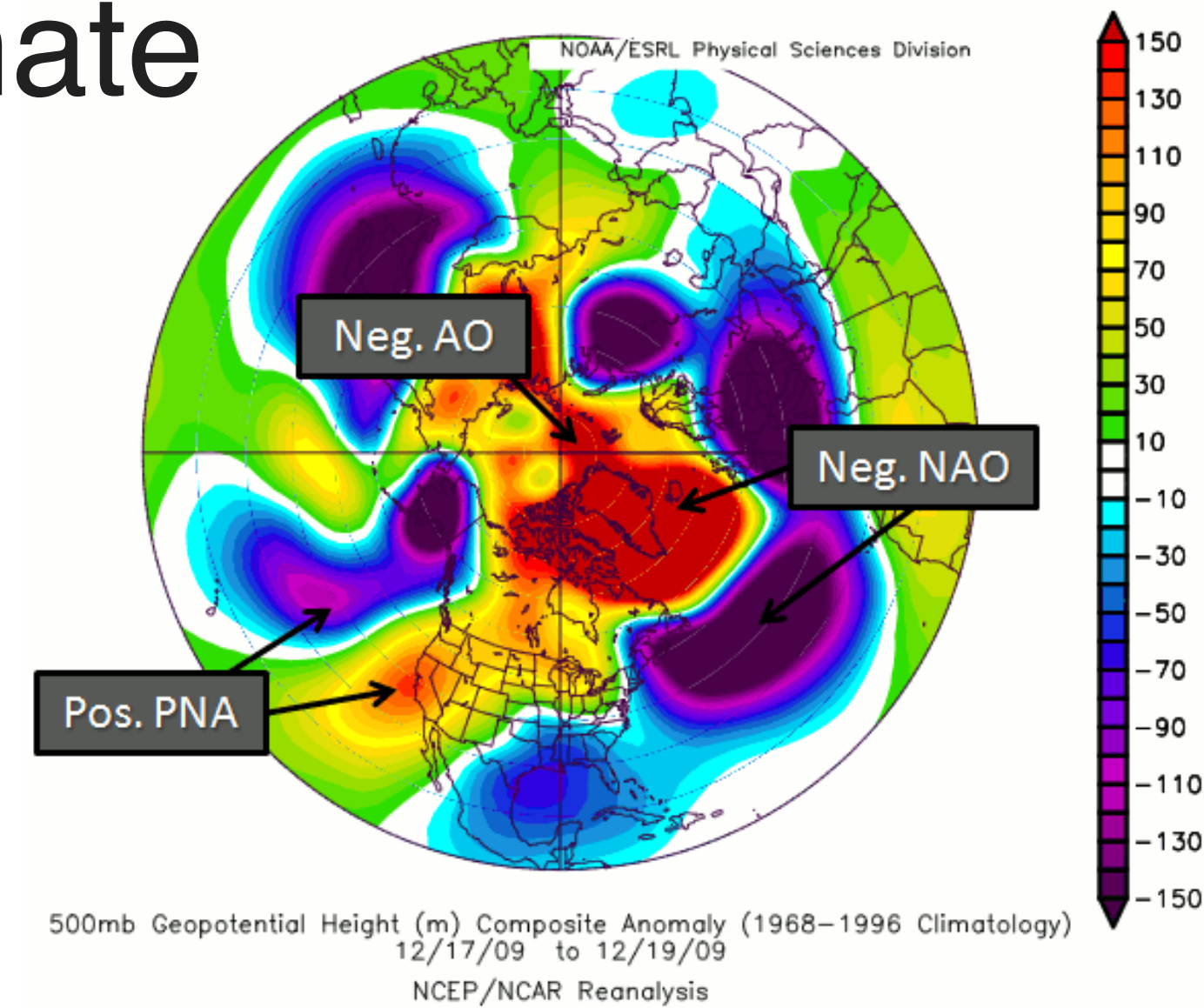
It simplifies, and can be the starting point for other purposes like data summarization.

Cluster analysis: useful for understanding/handling data

Medicine



Climate

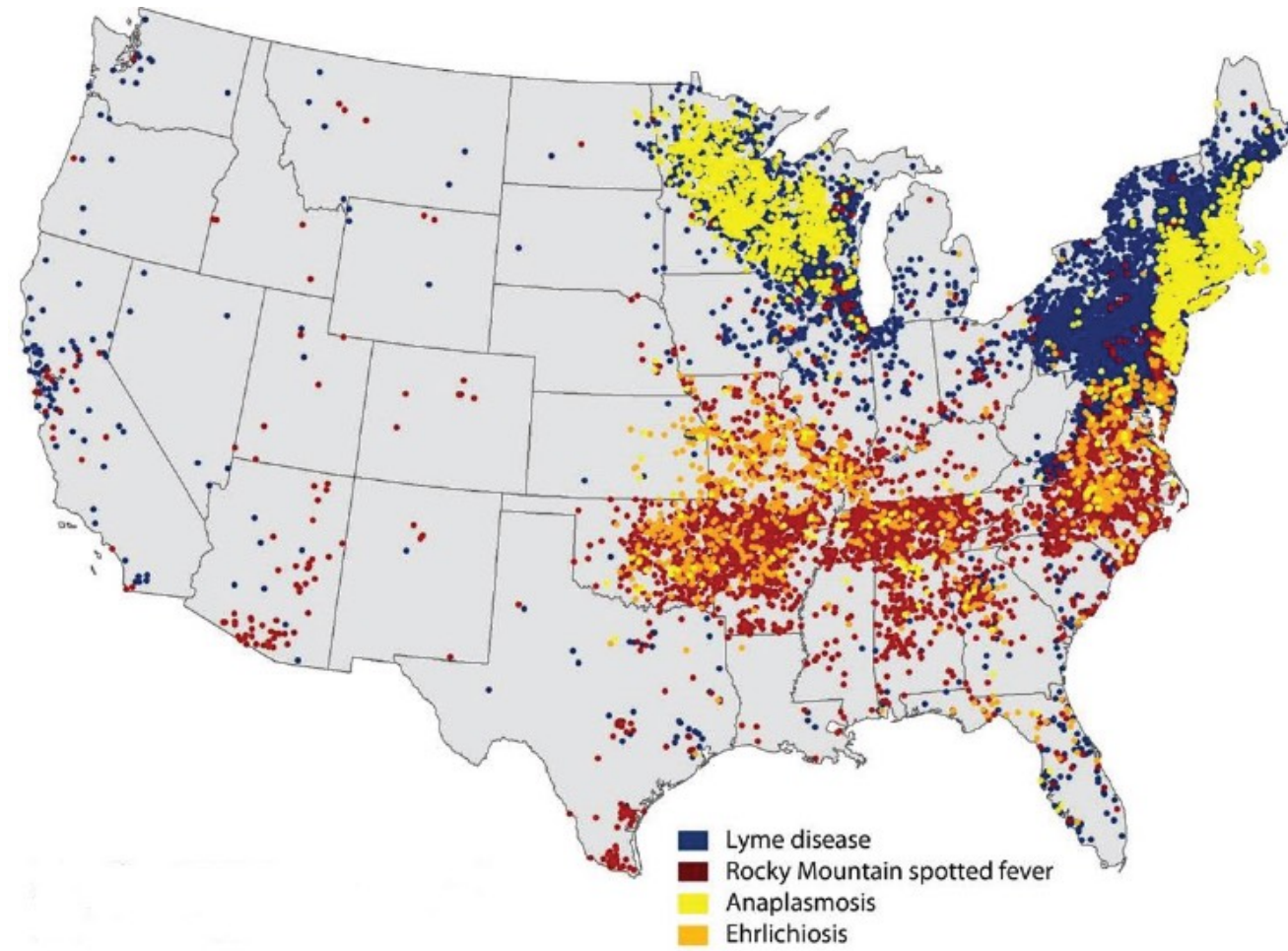


Business

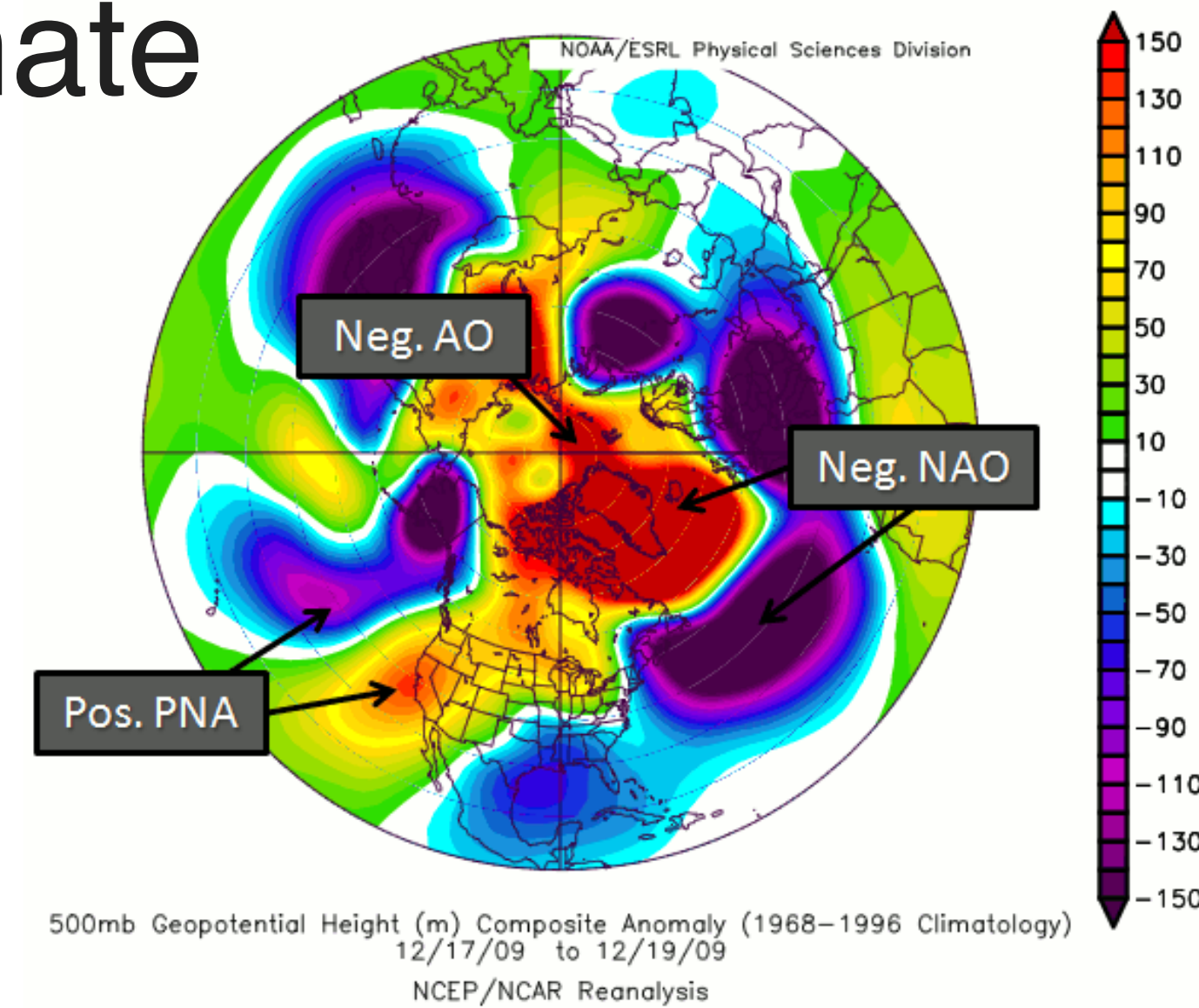
Segment customers for additional analysis
and to target for marketing activities

Cluster analysis: useful for understanding/handling data

Medicine



Climate



Business

Segment customers for additional analysis
and to target for marketing activities

Facebook

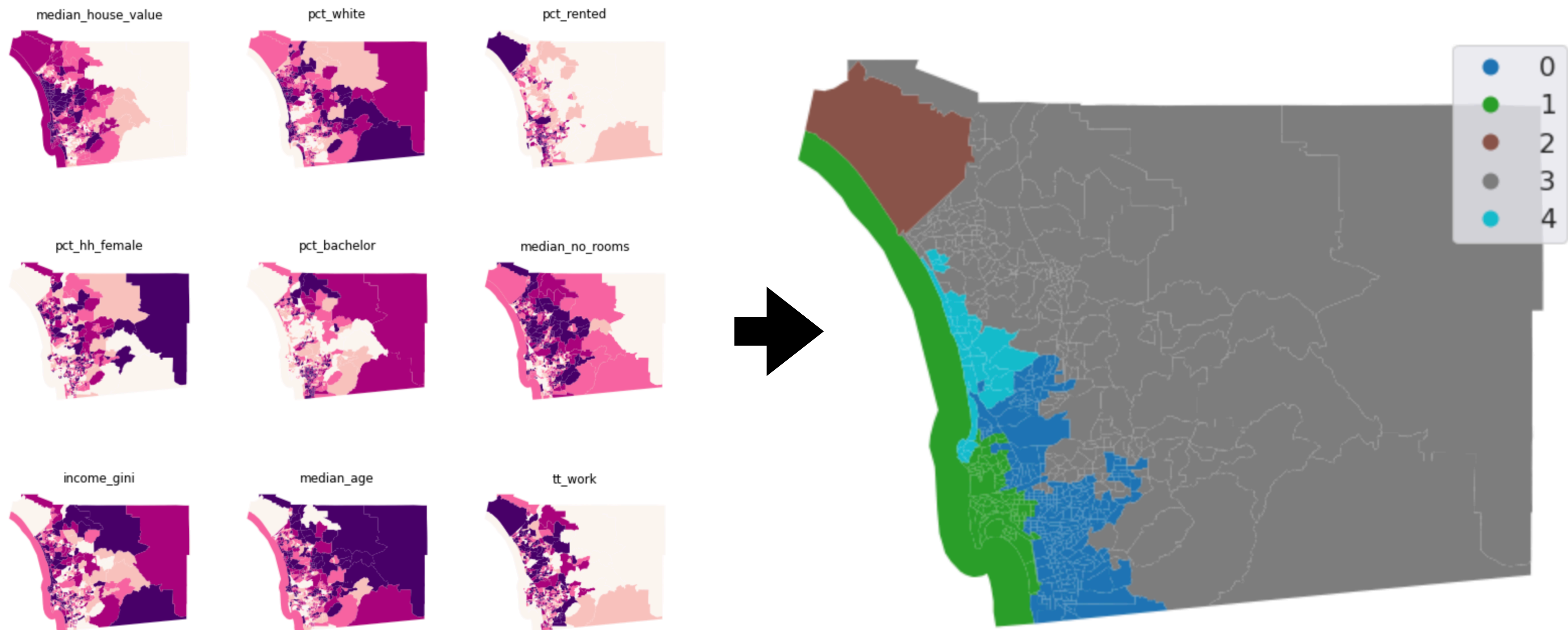
Facebook allowed advertisers to target 'Jew haters'

Embarrassing discovery that Facebook let advertisers target users
interested in antisemitic topics comes as the social network's ad
practices are under scrutiny

Clustering helps us make sense of complex spatial patterns

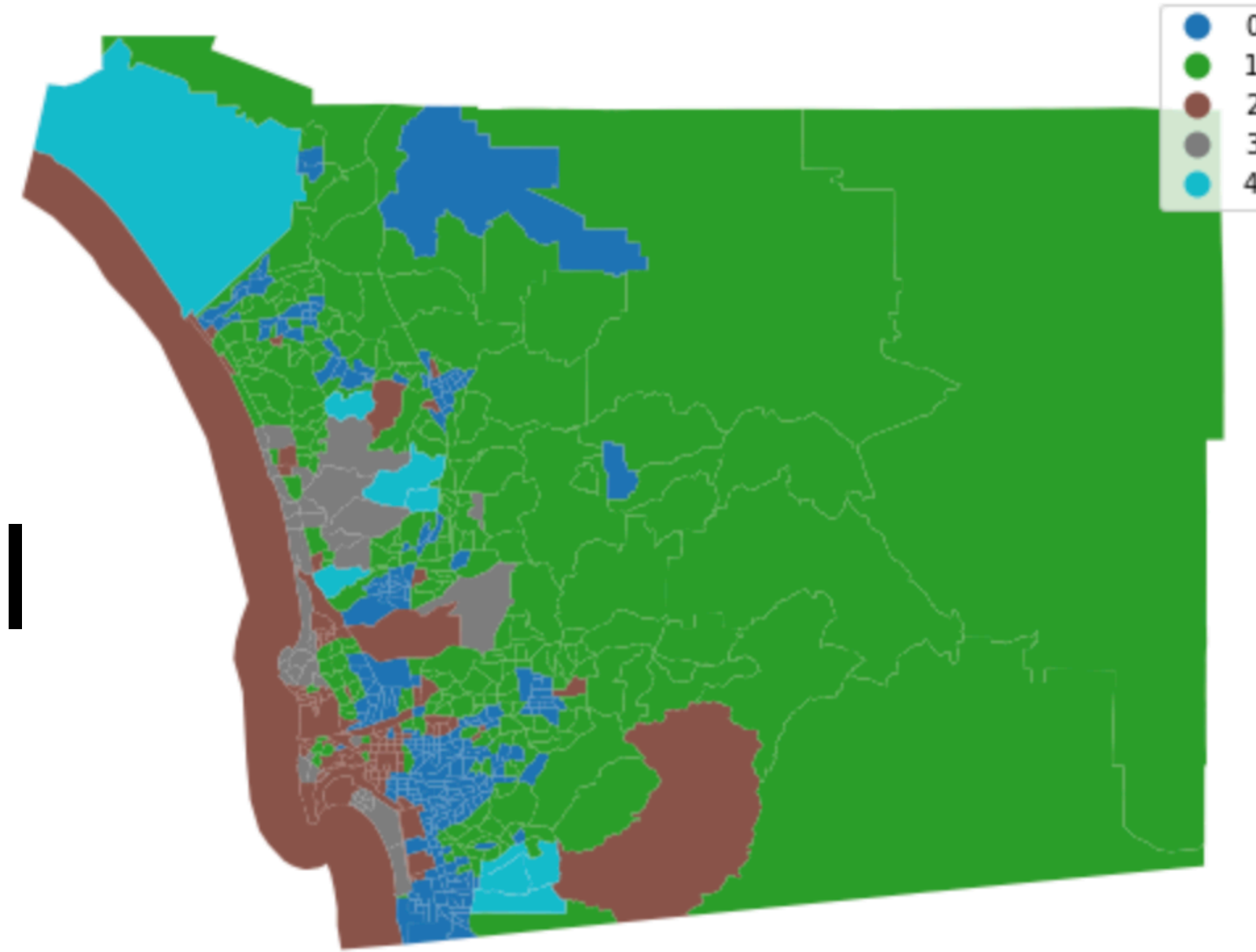


Clustering helps us make sense of complex spatial patterns

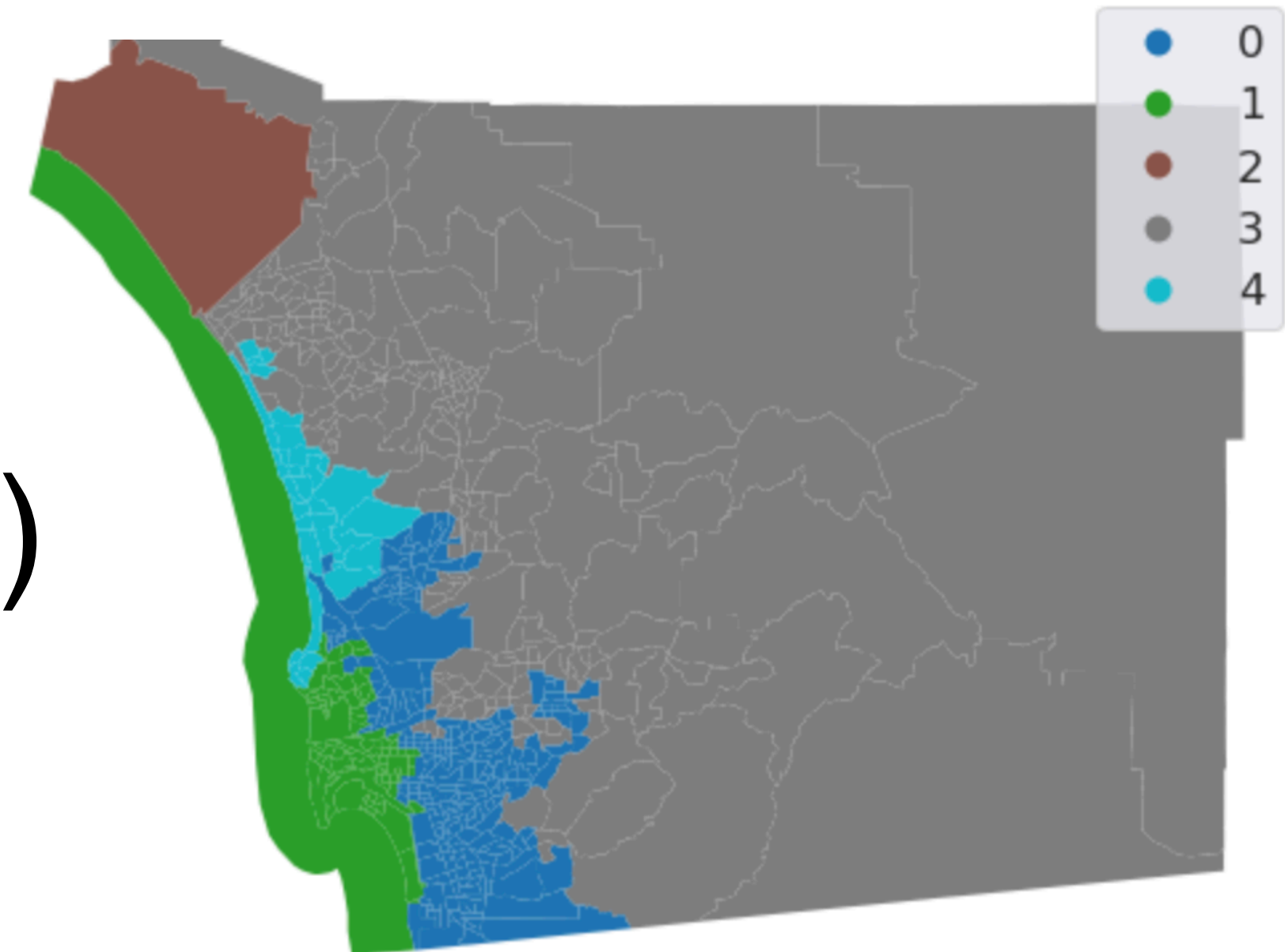


Types of clustering

Non-spatial

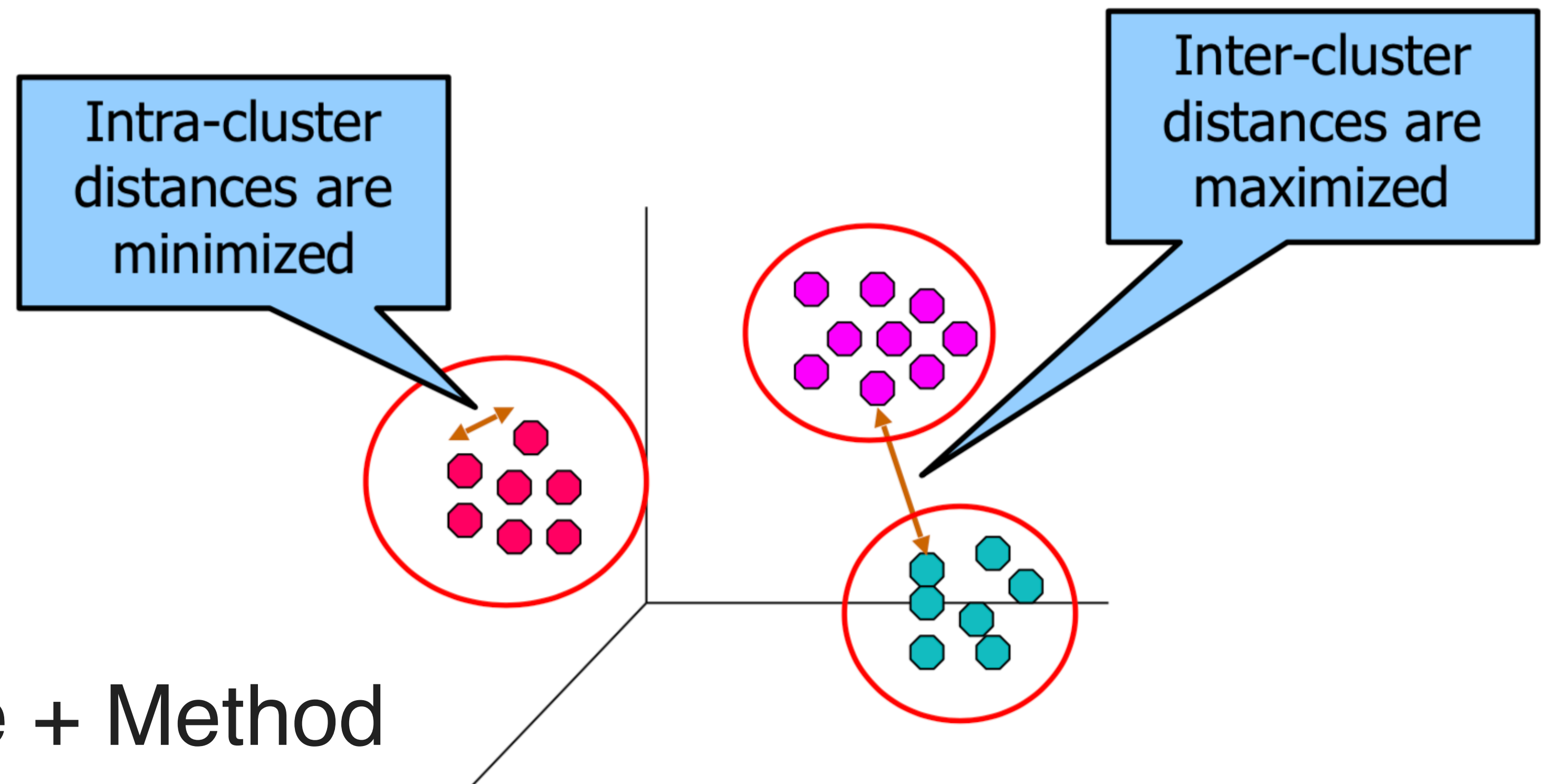


Spatial (Regionalization)



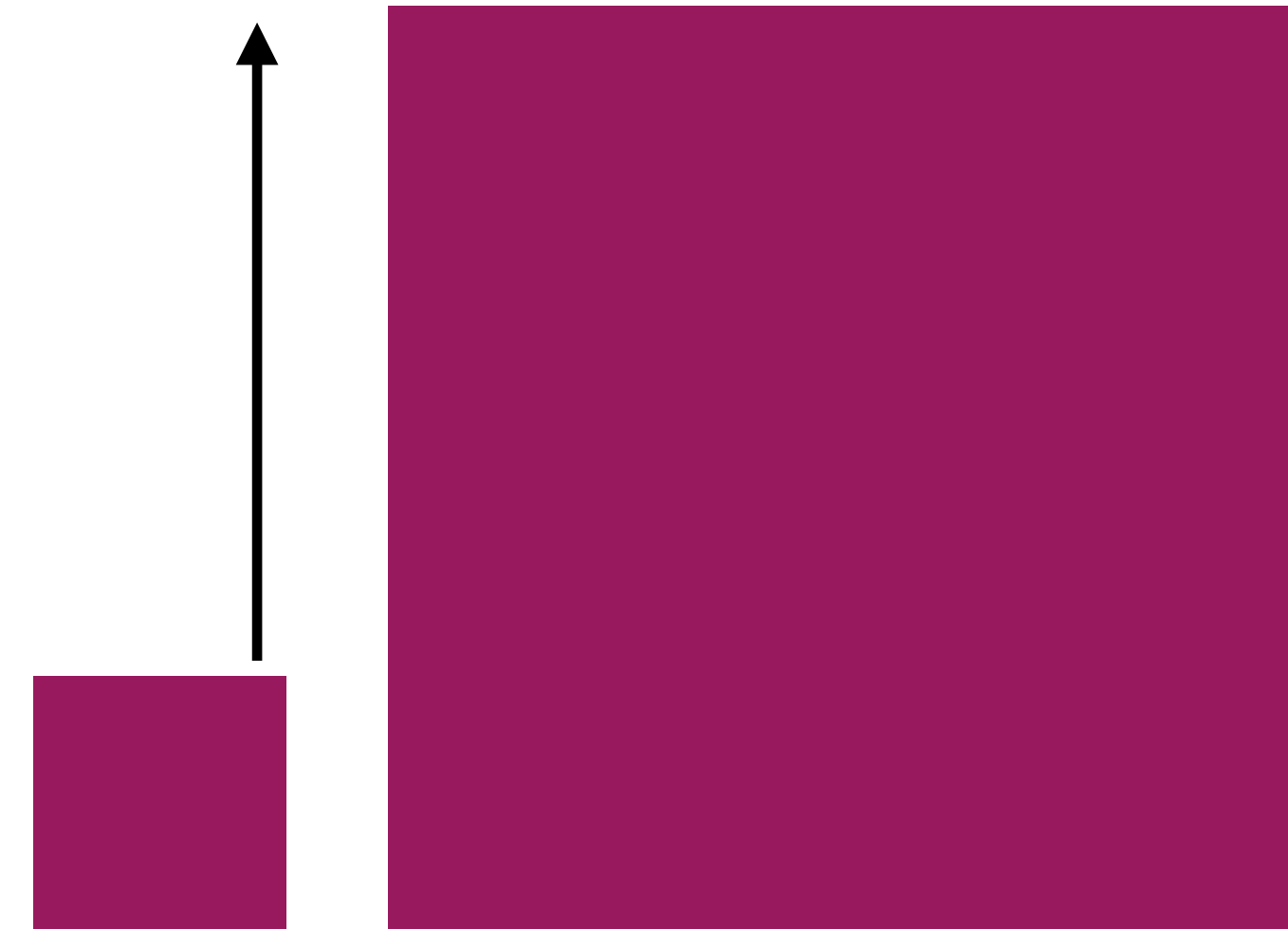
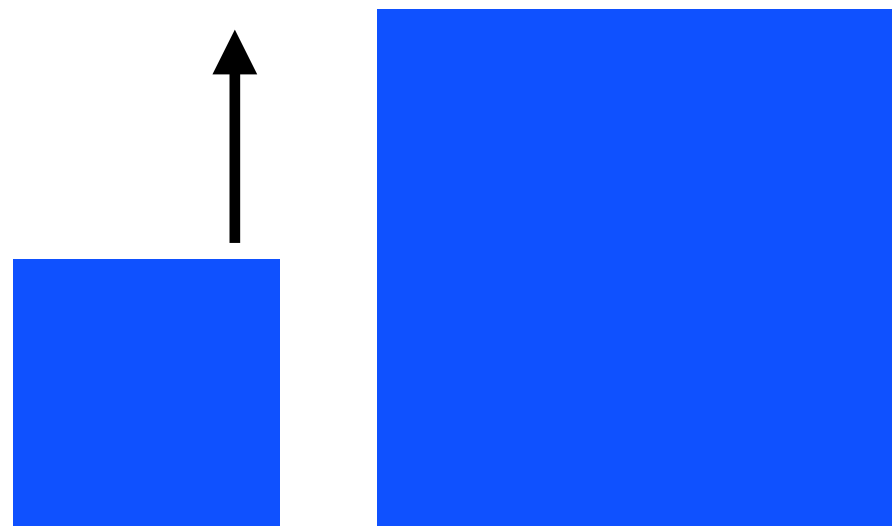
Cluster analysis: General concept

Finding groups of objects such that the objects in a group will be similar or related to one another and dissimilar or unrelated to the objects in other groups



Clustering = Distance + Method

Clustering requires scaled data



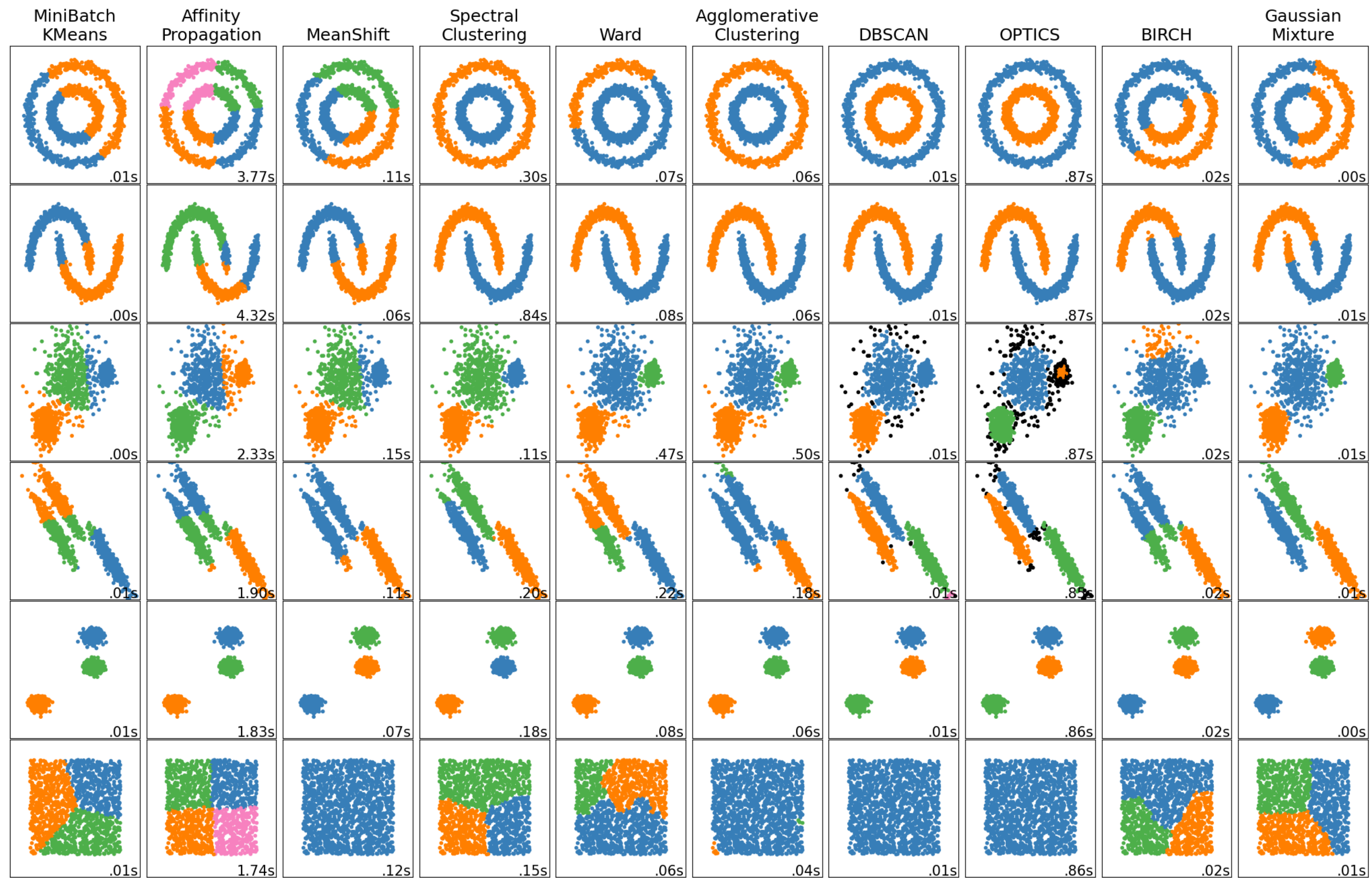
Clustering requires scaled data

Scale (standardize)

$$z = \frac{x_i - \bar{x}}{\sigma_x}$$

Robust Scale

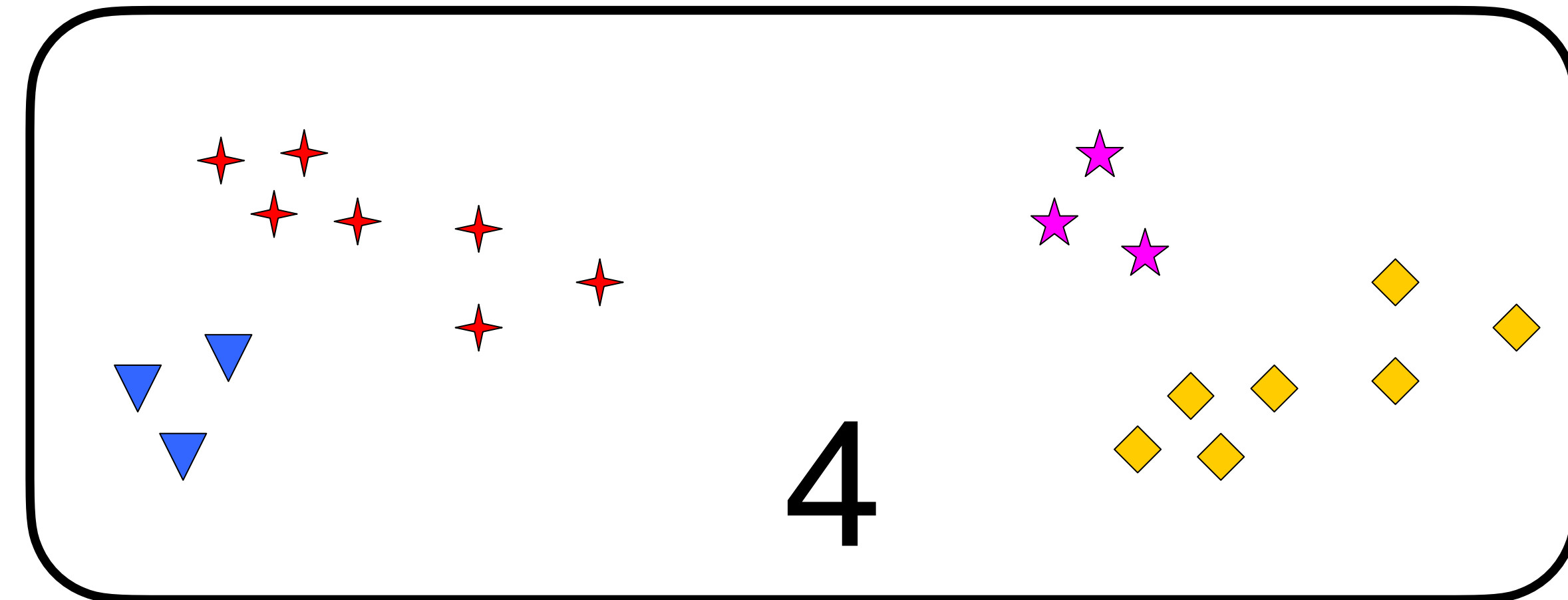
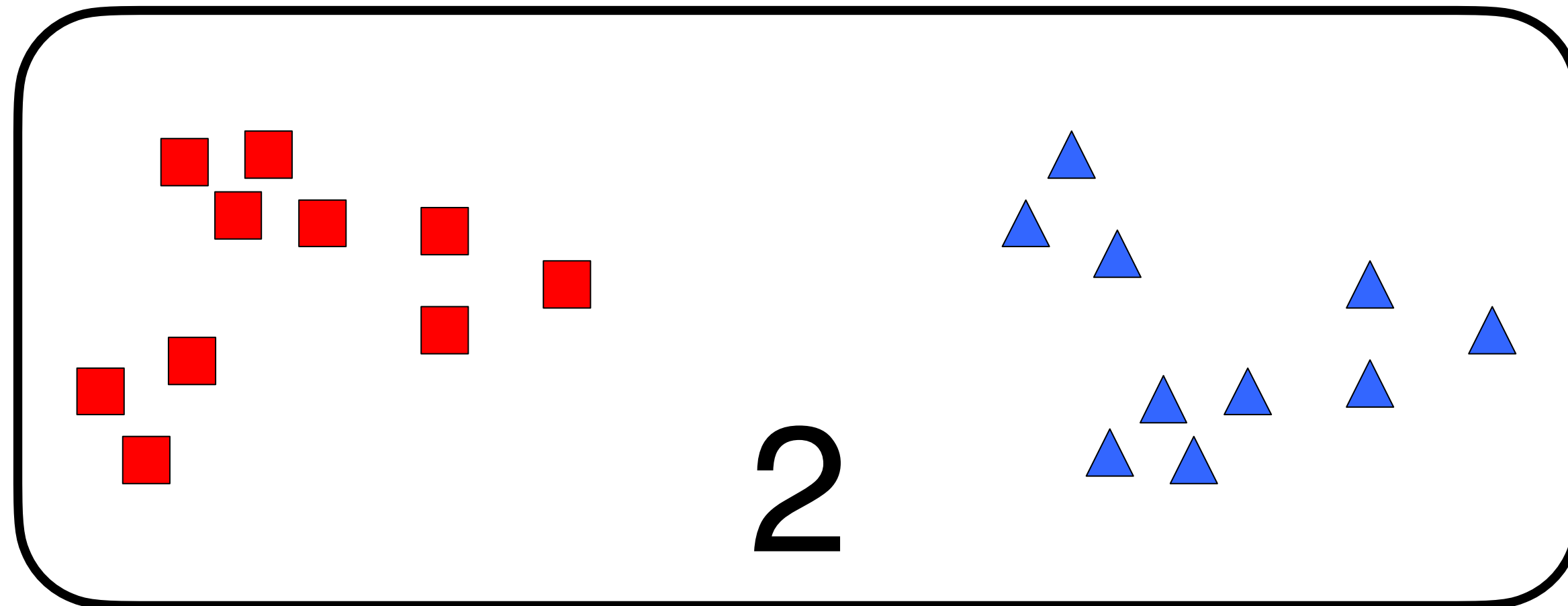
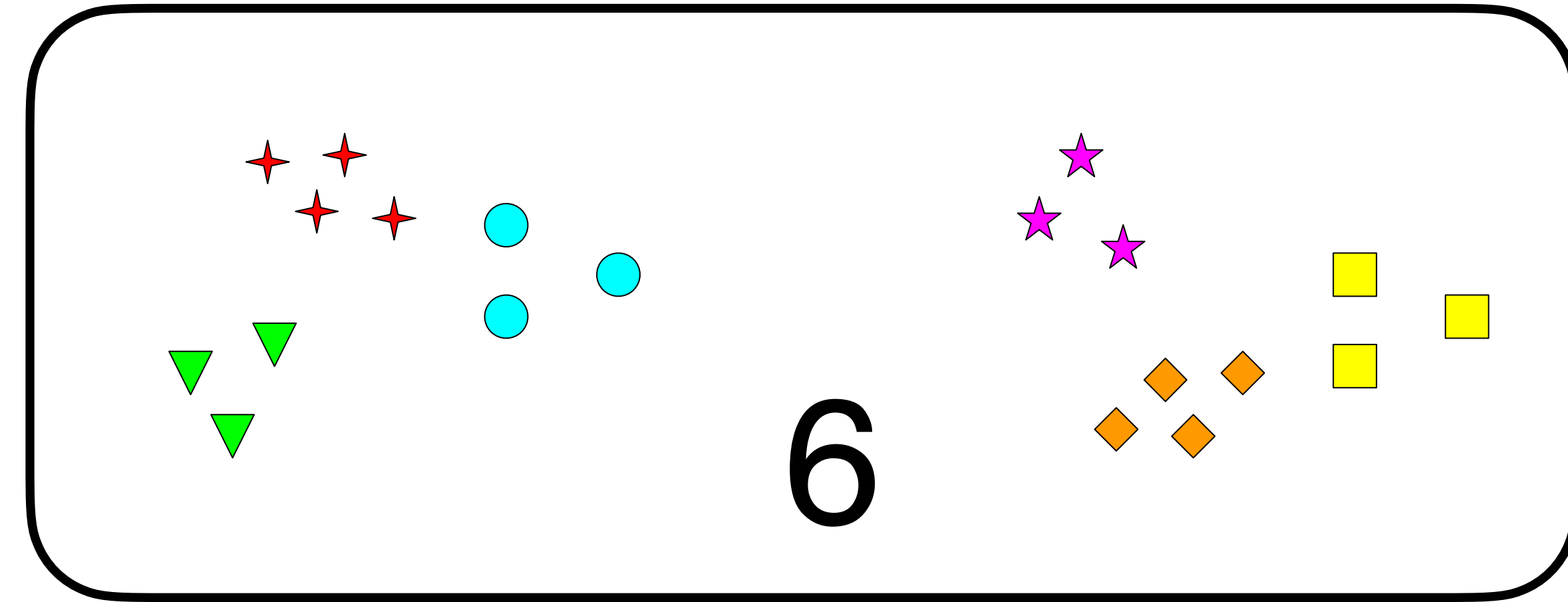
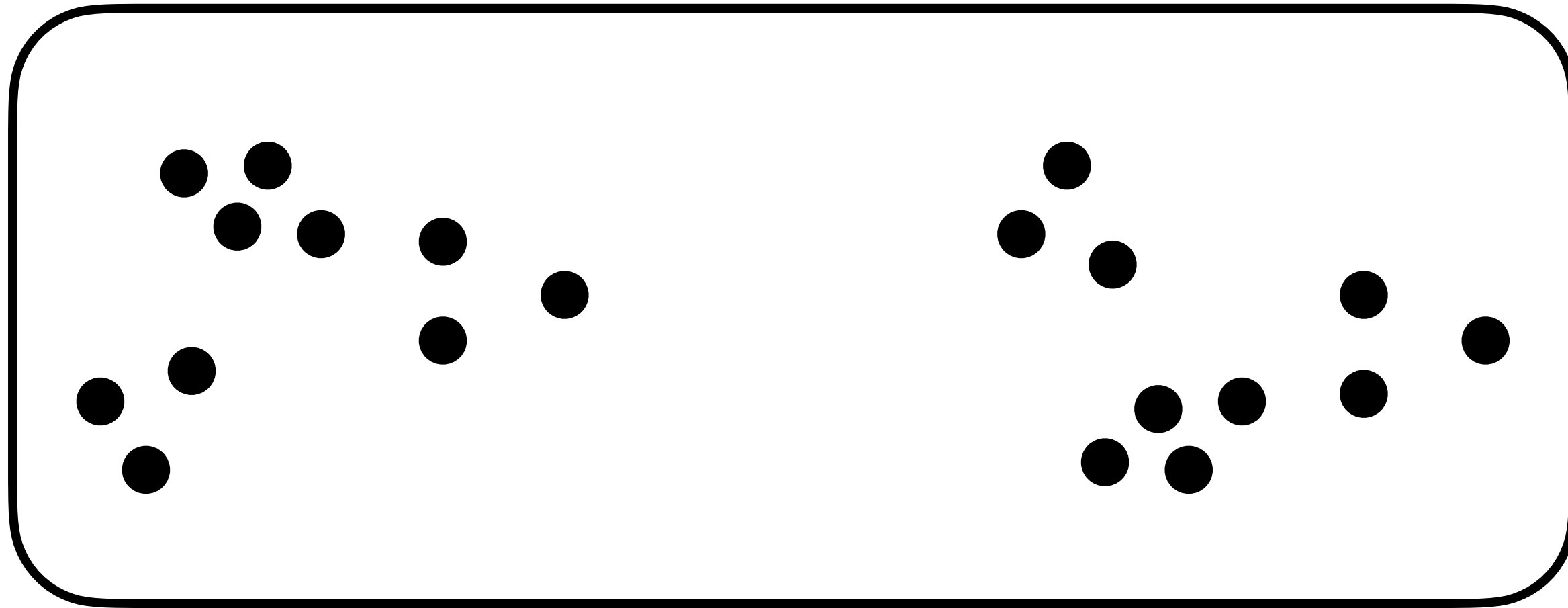
$$z = \frac{x_i - \tilde{x}}{[x]_{75} - [x]_{25}}$$



How many clusters are there?



Clustering is ambiguous

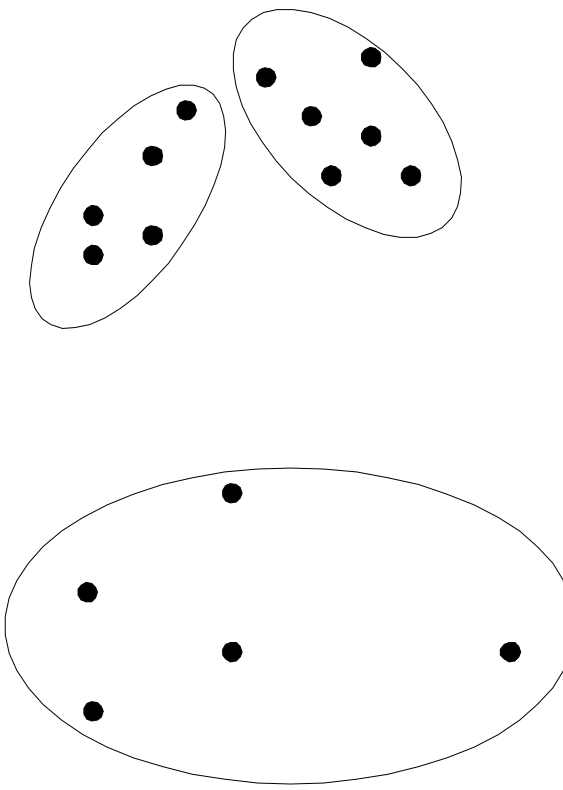


There is no clustering technique that is universally applicable in uncovering the variety of structures present in multidimensional data sets

Partitional vs hierarchical

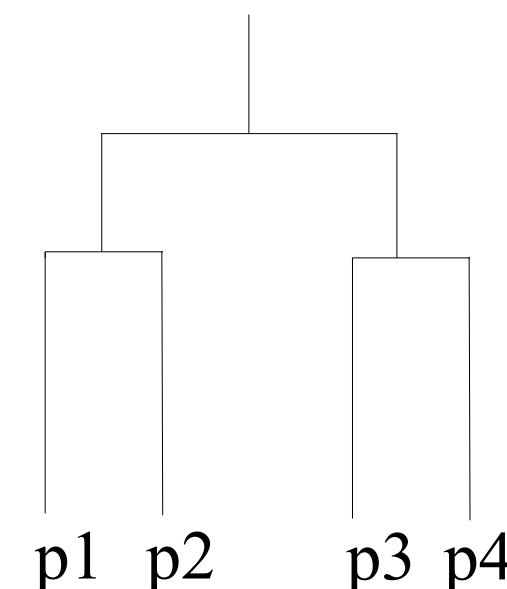
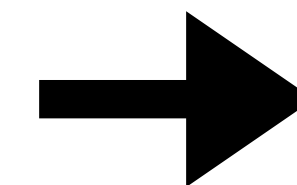
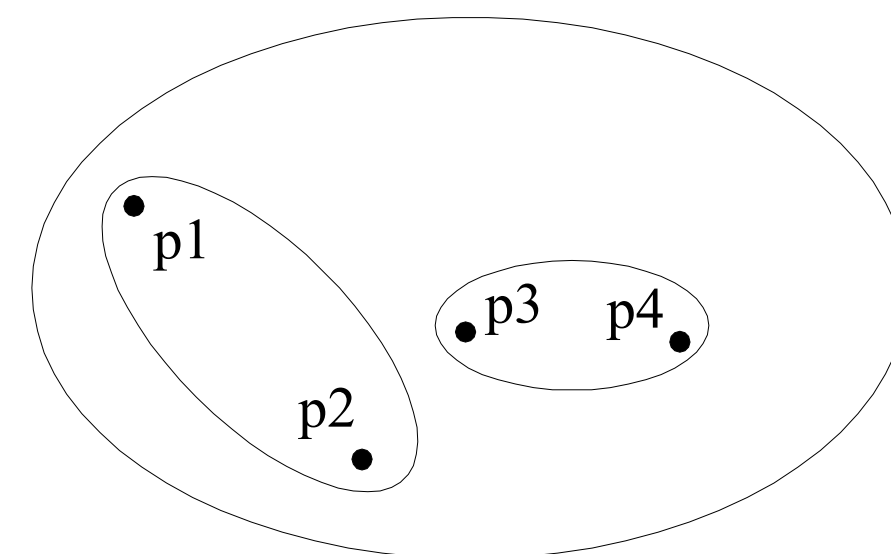
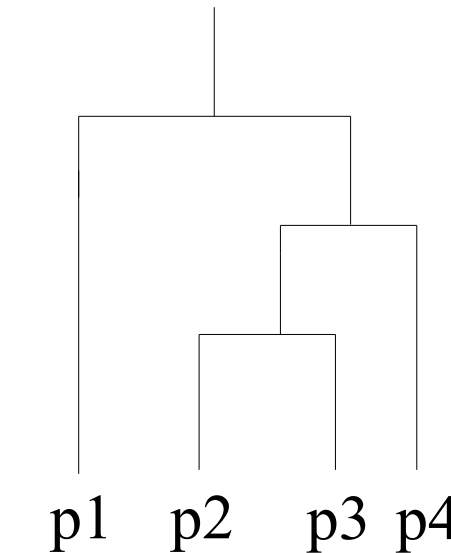
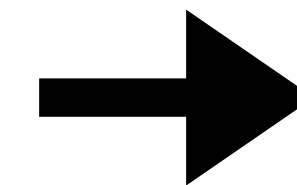
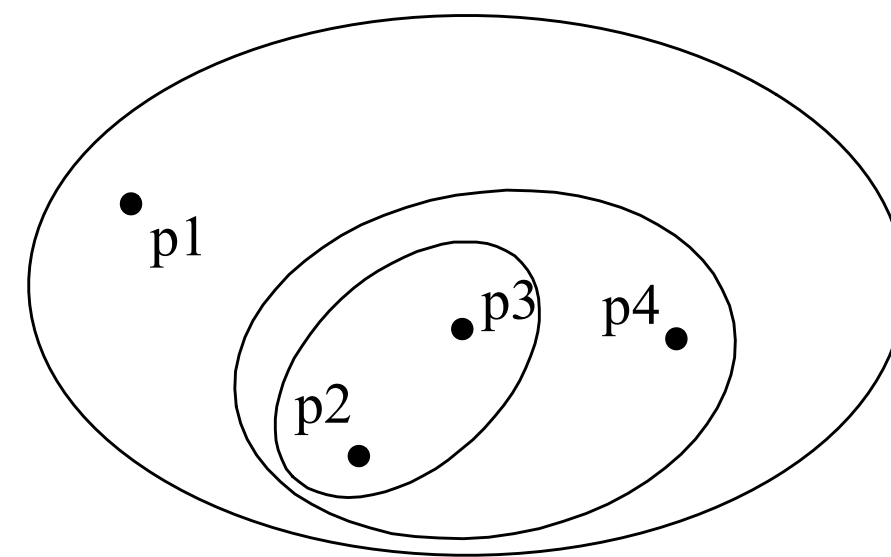
Partitional clustering

A division into non-overlapping subsets. Each data object is in exactly one subset.



Hierarchical clustering

A set of nested clusters organized as a tree



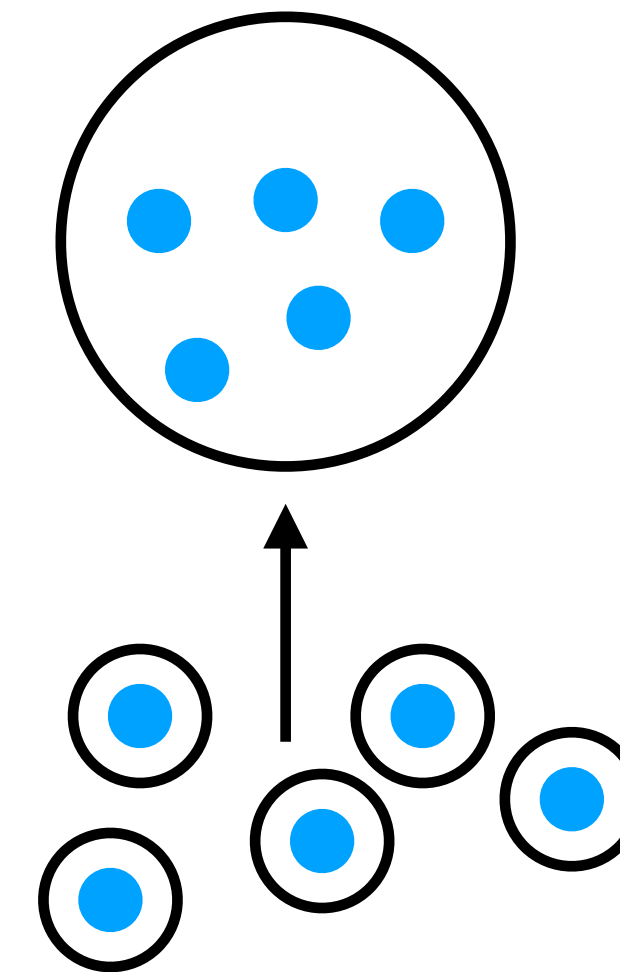
Hierarchical clustering

Agglomerative clustering (bottom-up)

Start with points being individual clusters

At each step merge the closest pair of clusters.

Needs: Similarity measure

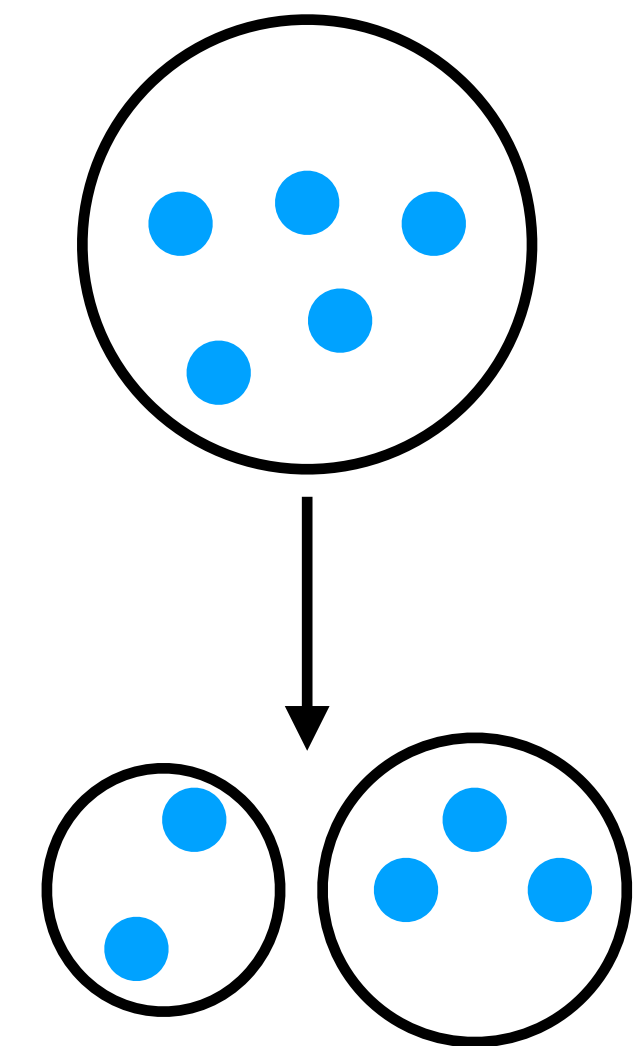


Divisive clustering (top-down)

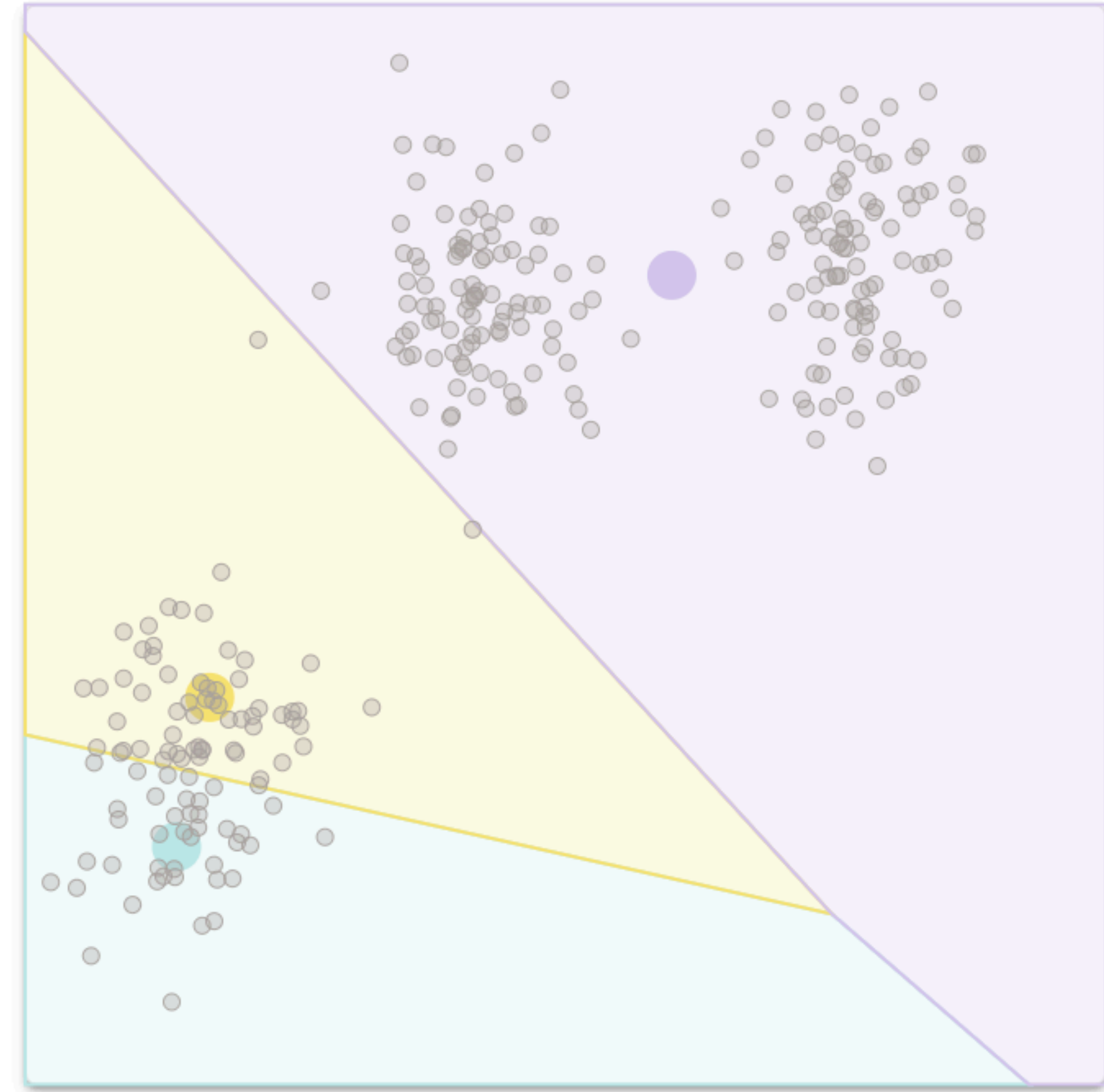
Start with one all-inclusive cluster.

At each step split a cluster until clusters of single points remain.

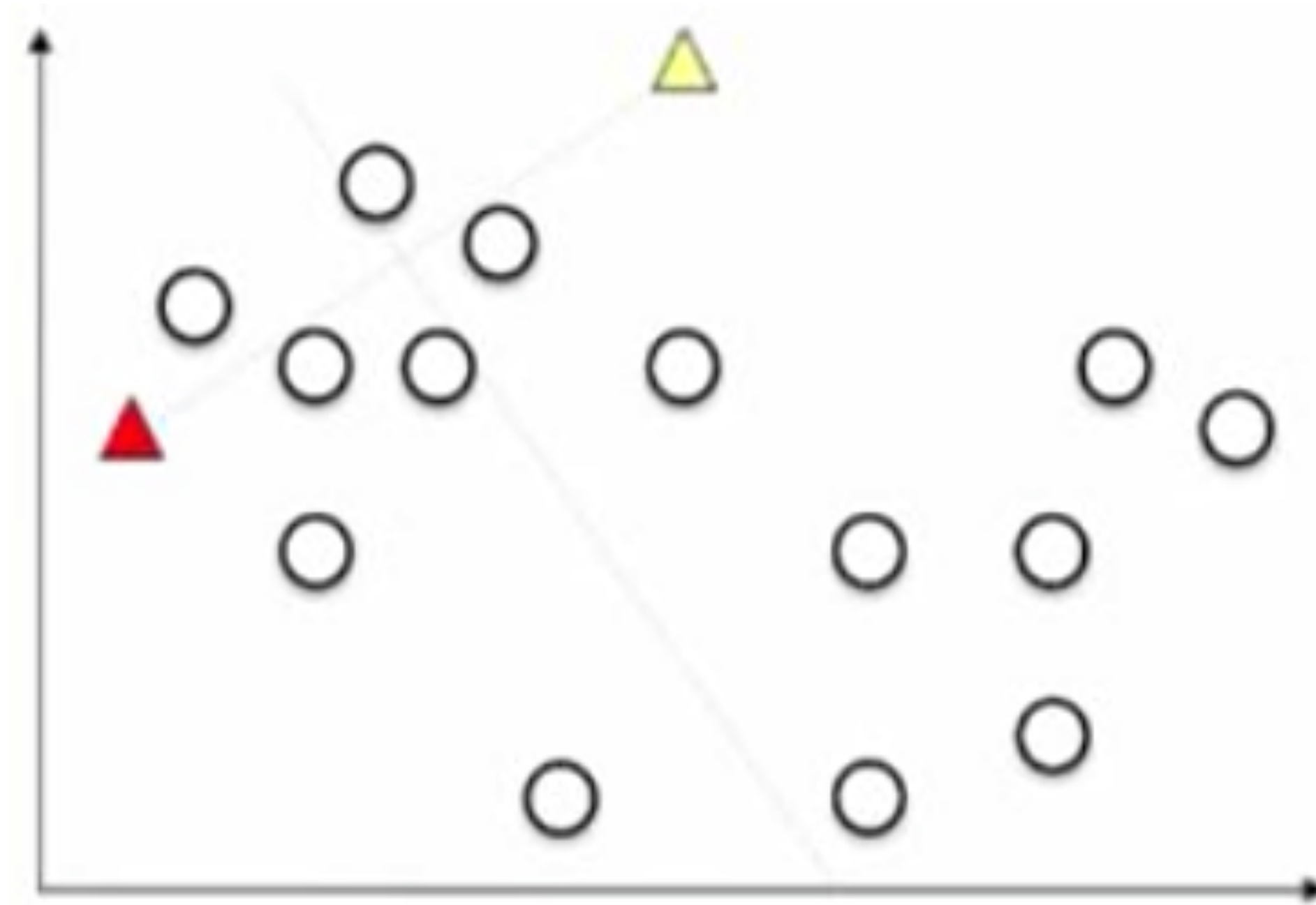
Needs: Way to decide a split



K-means

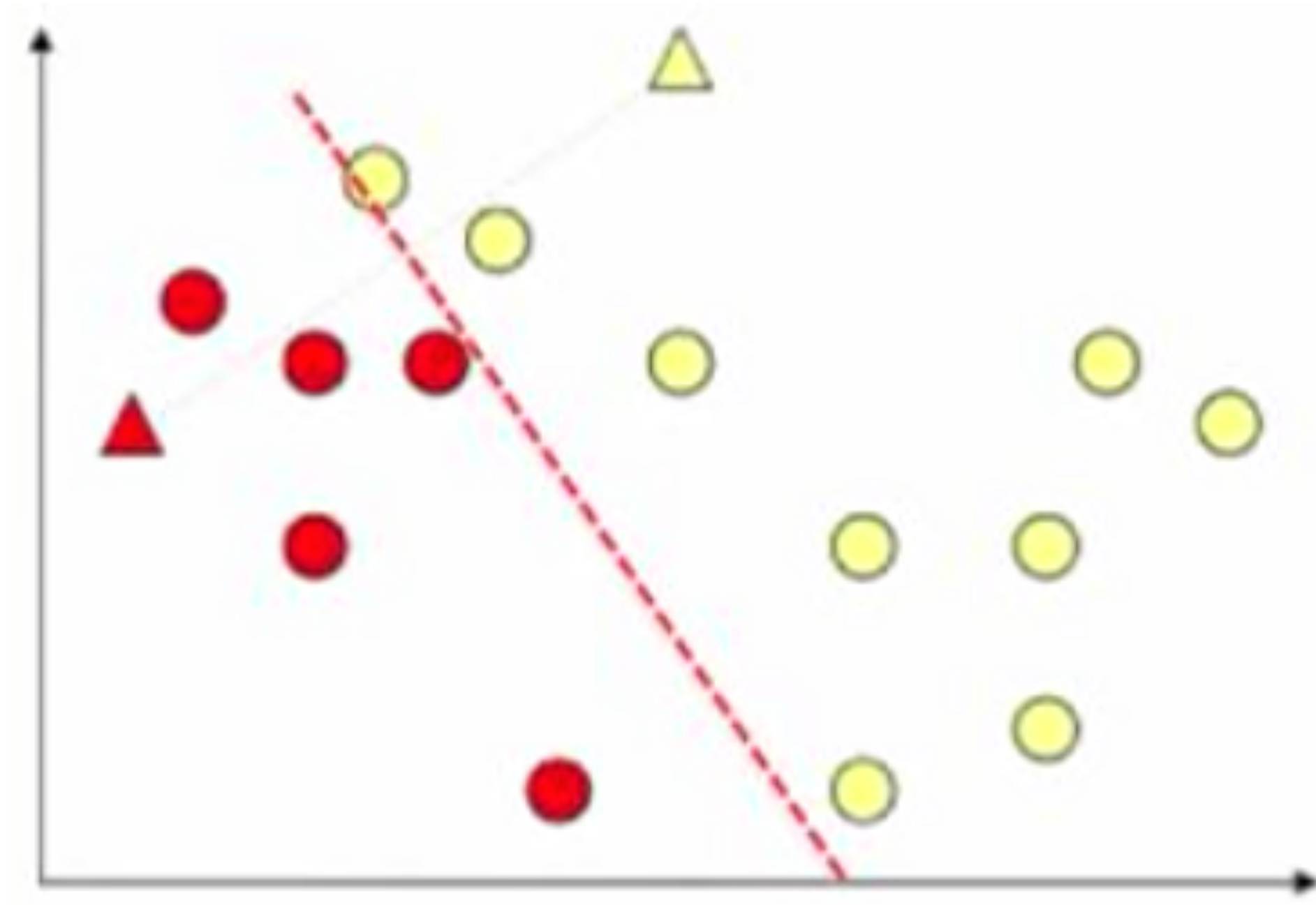


<https://k-means-explorable.vercel.app/>



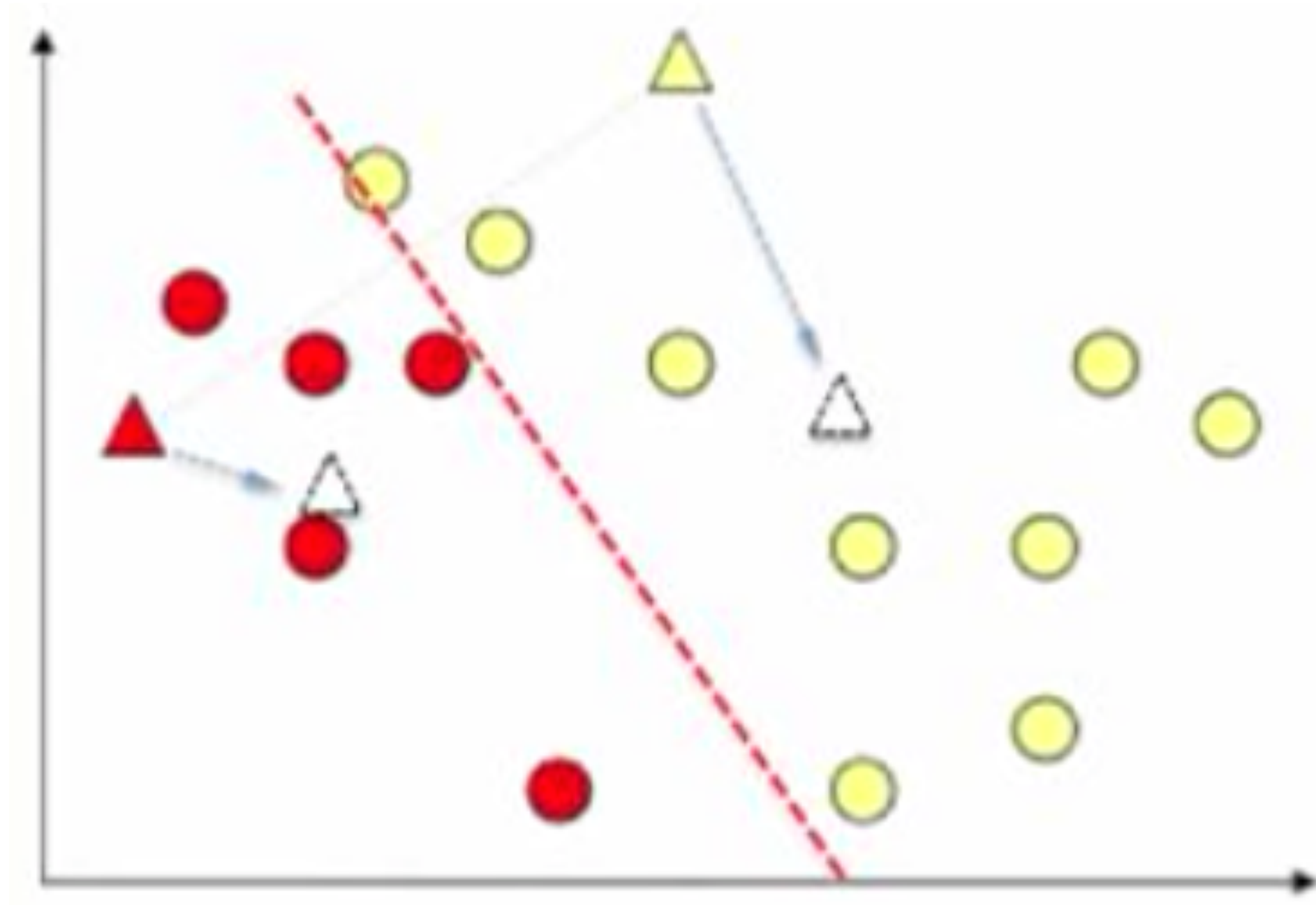
Algorithm 8.1 Basic K-means algorithm.

- 1: Select K points as initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning each point to its closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** Centroids do not change.
-



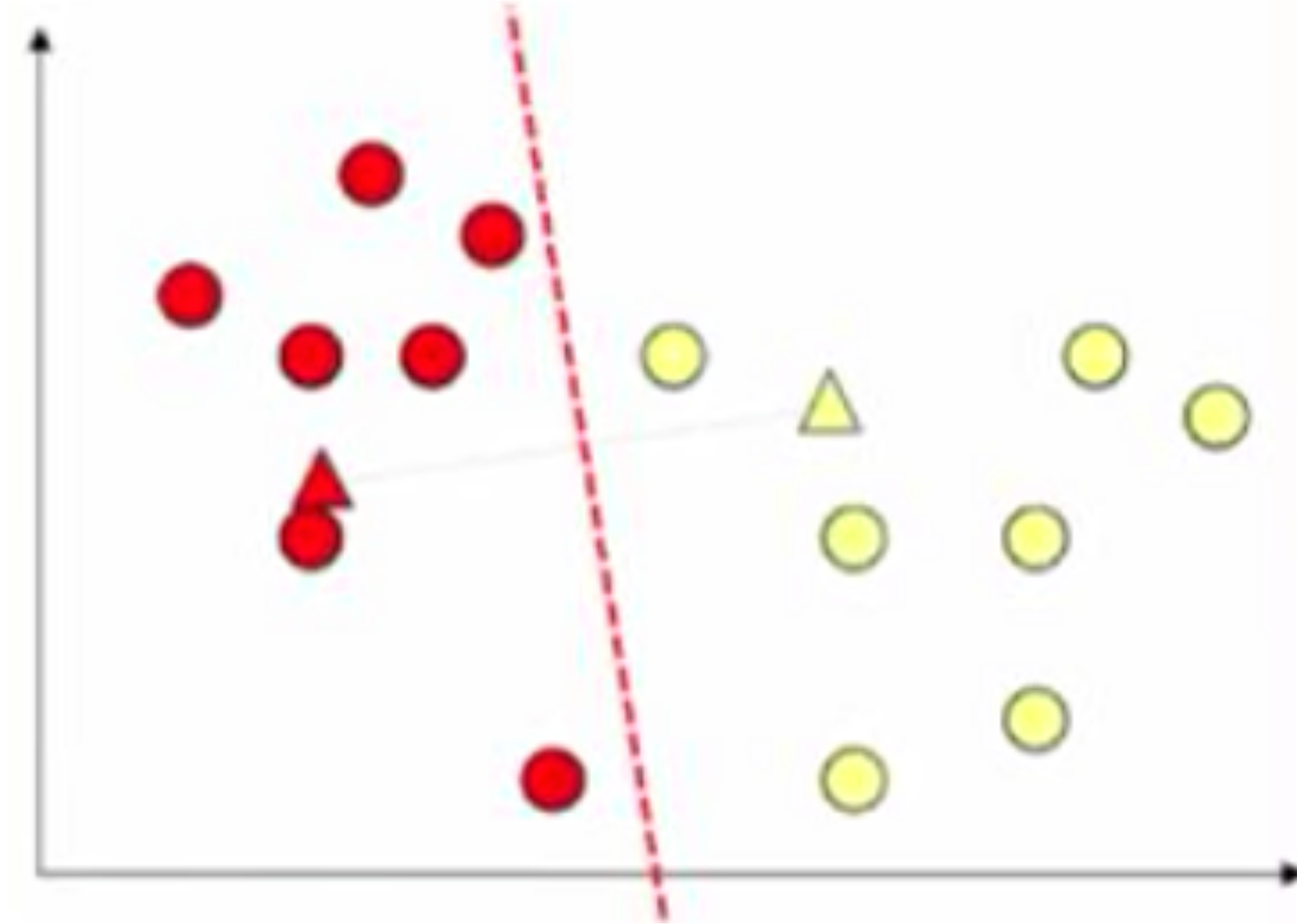
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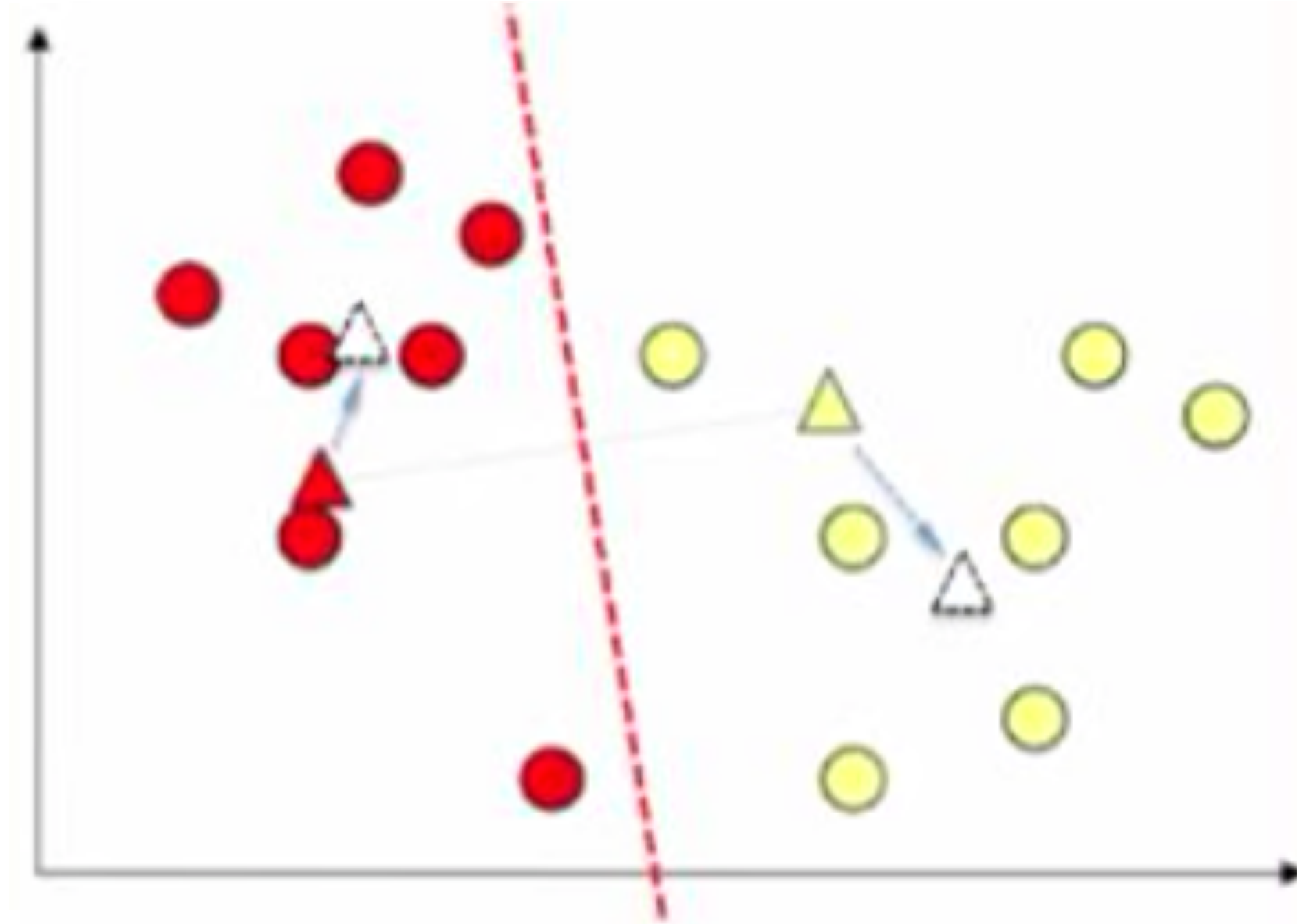
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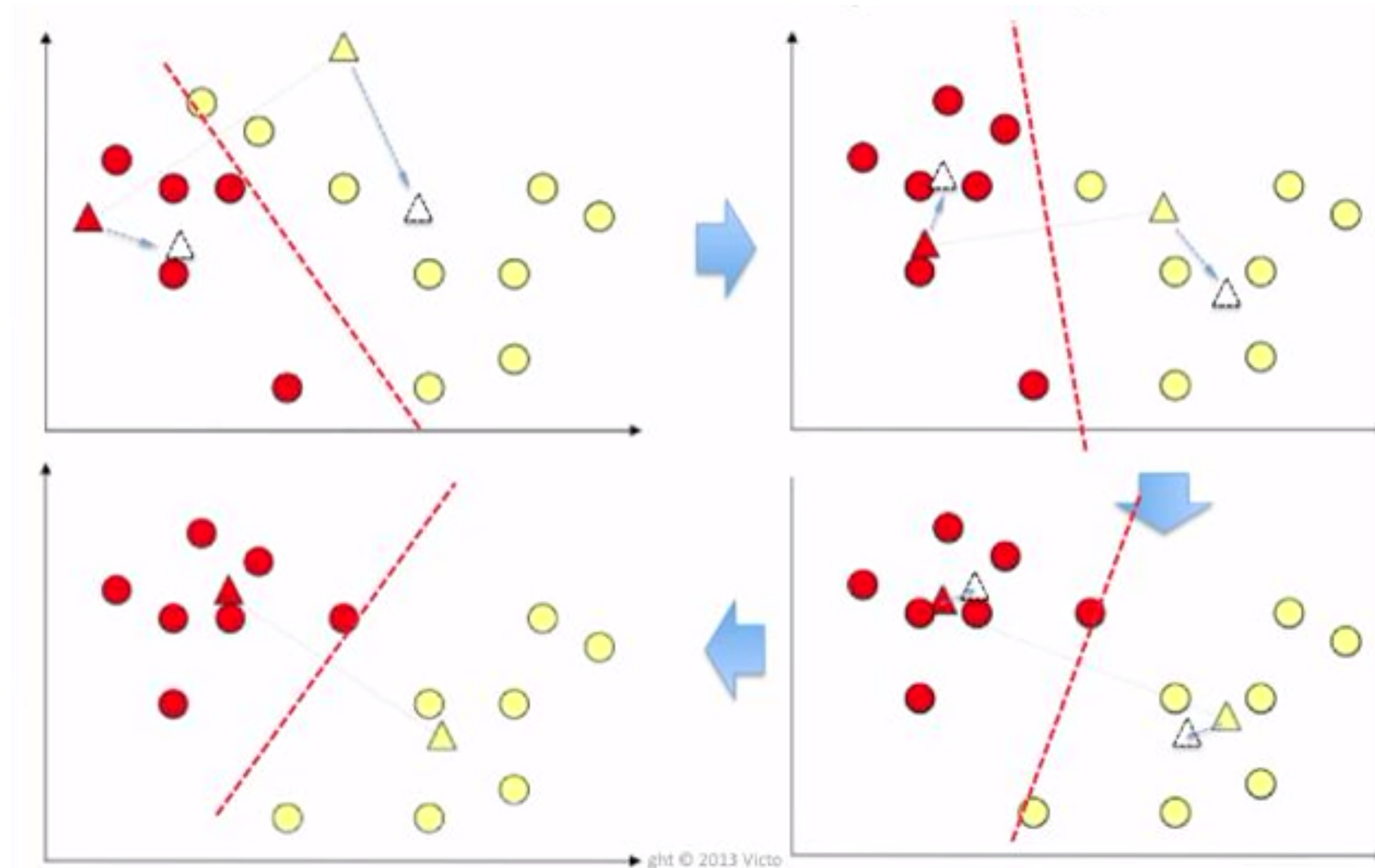
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Algorithm 8.1 Basic K-means algorithm.

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-

K-means: Math background

We want to minimize the sum of squared errors (SSE)

$$SSE = \sum_{i=1}^K \sum_{\mathbf{x} \in C_i} dist(\mathbf{c}_i, \mathbf{x})^2$$

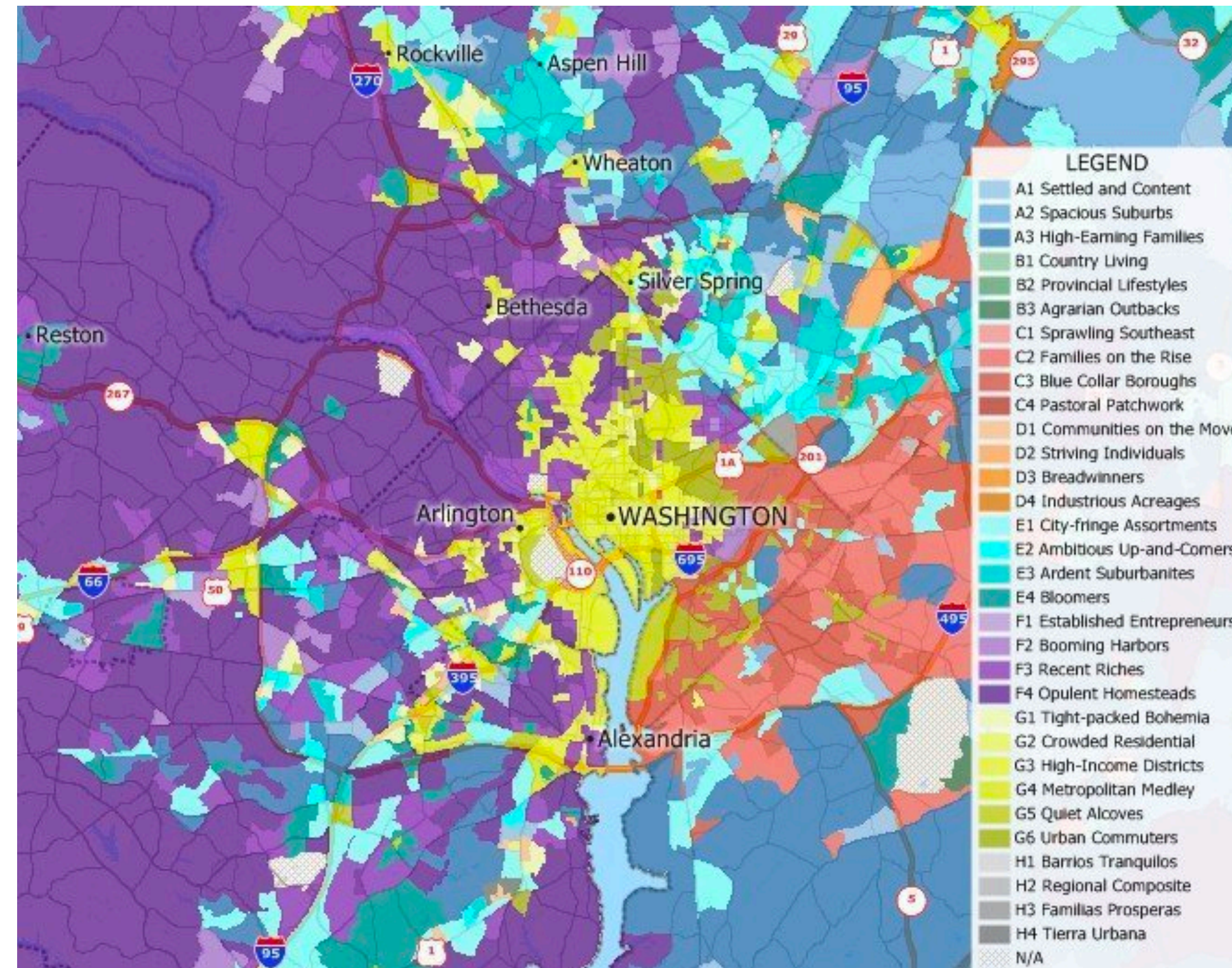
The centroid of the i th cluster is just the mean of the points

$$\mathbf{c}_i = \frac{1}{m_i} \sum_{\mathbf{x} \in C_i} \mathbf{x}$$

Distance can be Euclidian, Manhattan, Minkowski, ..

Geodemographic analysis

Geodemographic analysis aims to identify similar “neighborhoods” based on people and place characteristics



Does our data set even have clusters?

A clustering algorithm will do its job and give us clusters.

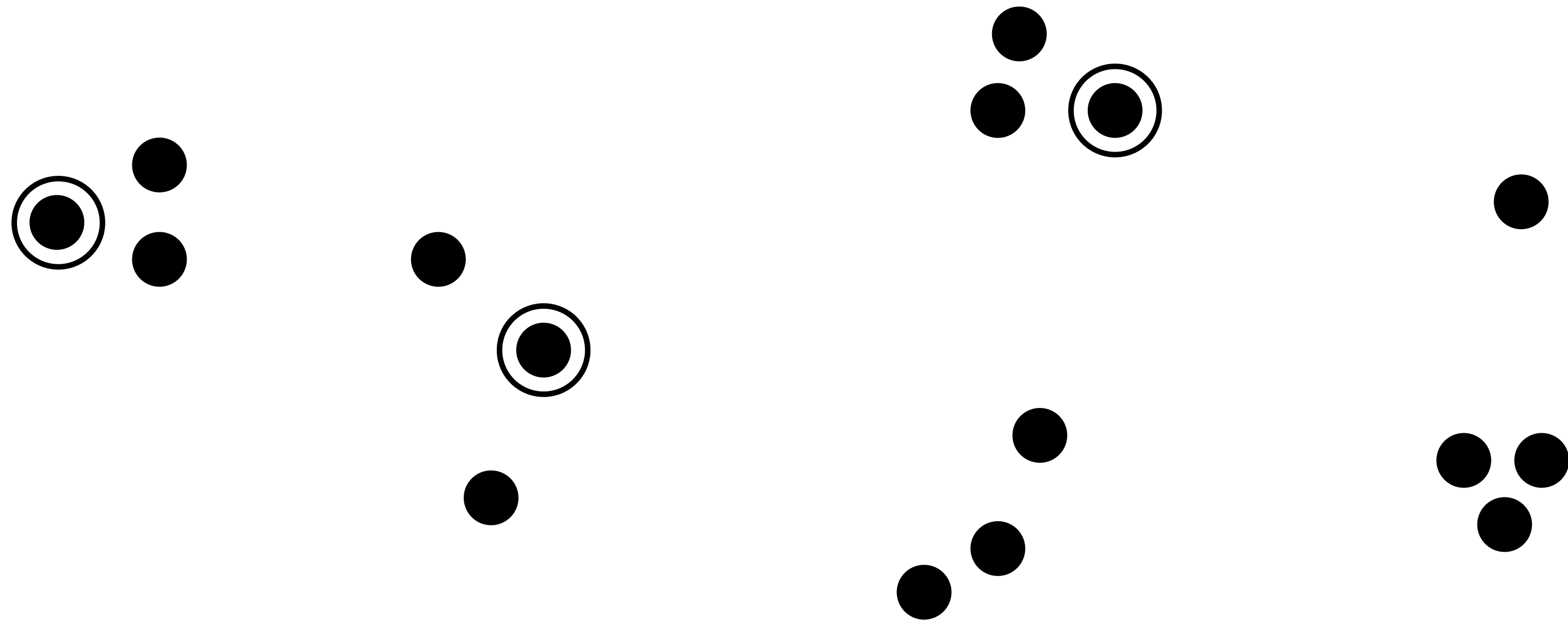
But does it make sense?

The Hopkins statistic H measures clustering tendency

A bit like Moran's I , but usually used in a non-spatial context

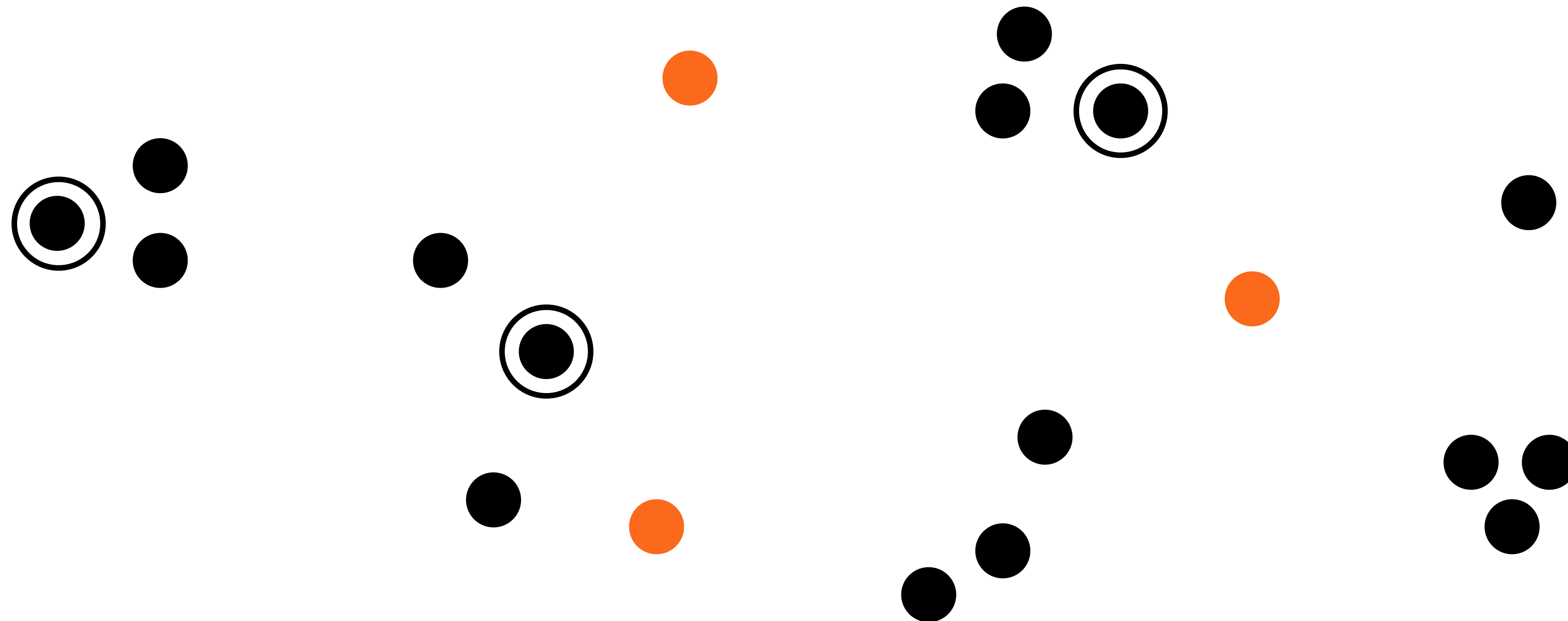
The Hopkins statistic H measures clustering tendency

1) Sample m data points out of $n \gg m$



The Hopkins statistic H measures clustering tendency

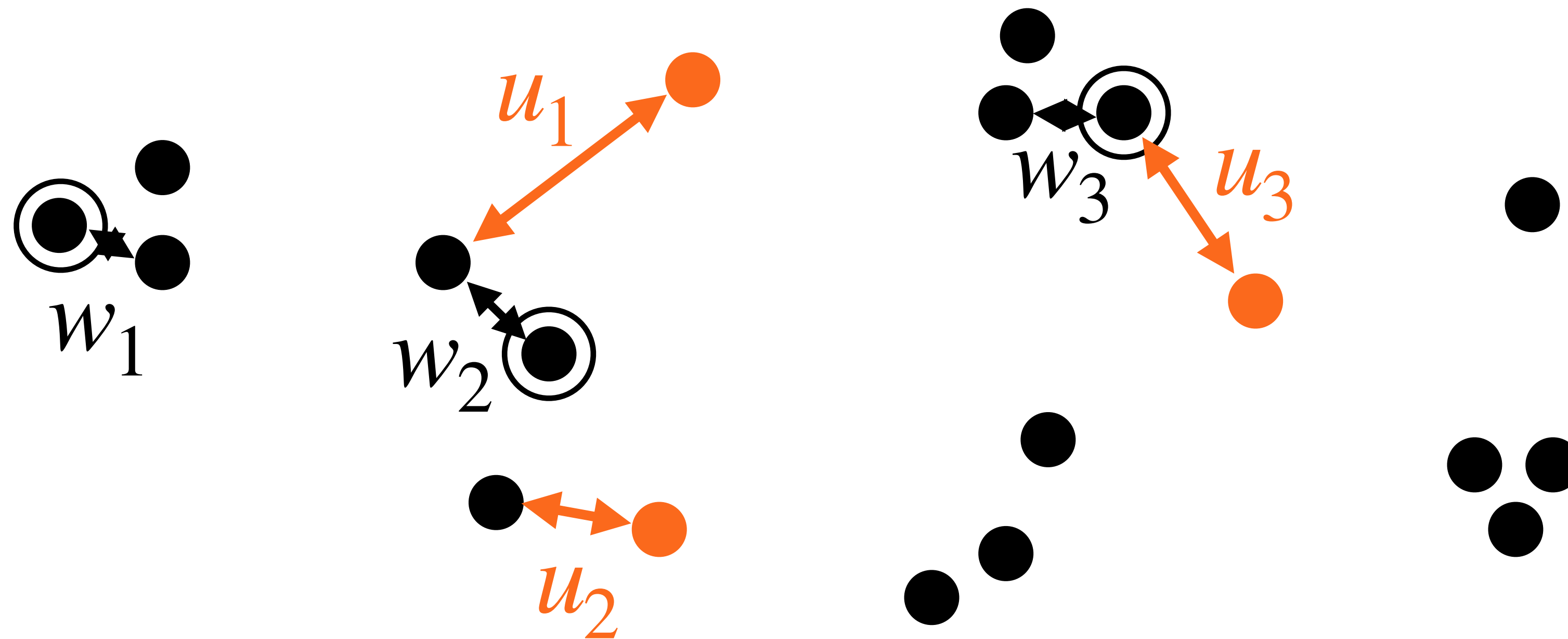
- 1) Sample m data points out of $n \gg m$
- 2) Generate m random synthetic data points



The Hopkins statistic H measures clustering tendency

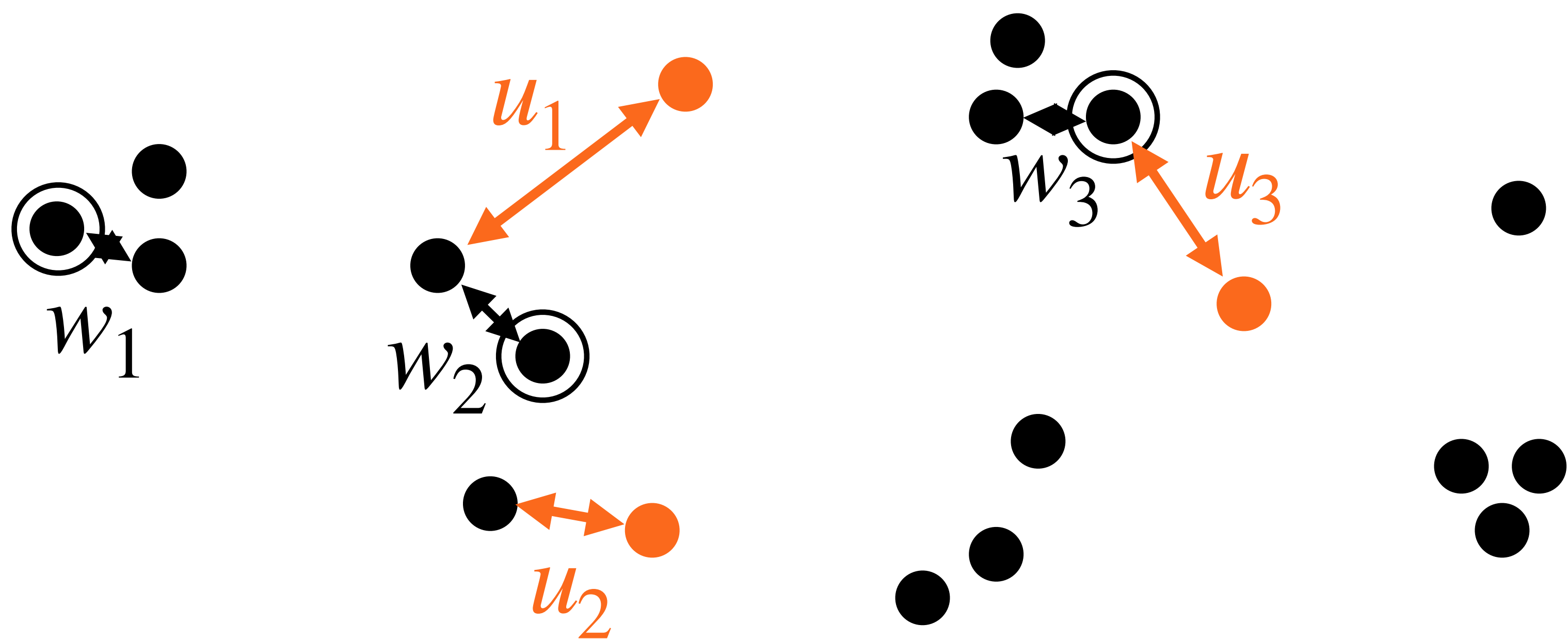
- 1) Sample m data points out of $n \gg m$
- 2) Generate m random synthetic data points

For both sets, find the distance to the nearest neighbors in the original data set.



The Hopkins statistic H measures clustering tendency

$$H = \frac{\sum_{i=1}^m u_i}{\sum_{i=1}^m w_i + \sum_{i=1}^m u_i}$$



The Hopkins statistic H measures clustering tendency

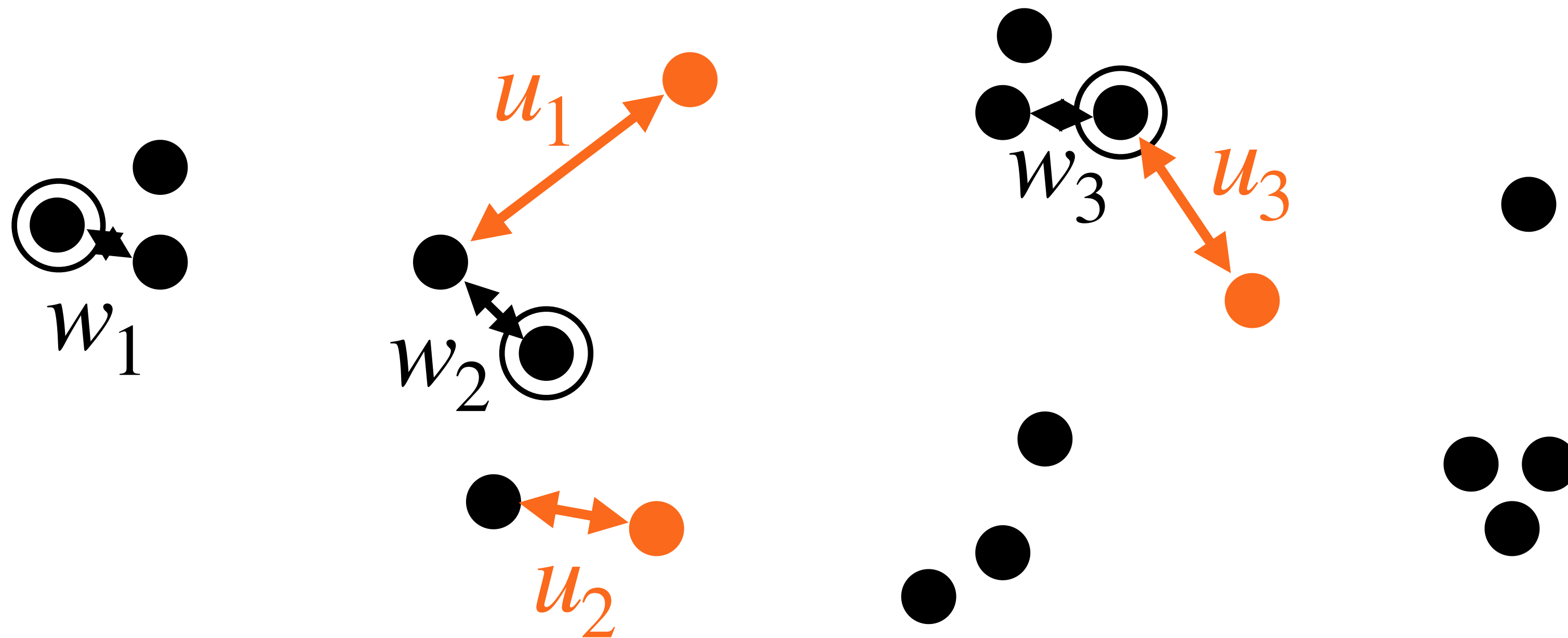
$$H = \frac{\sum_{i=1}^m u_i}{\sum_{i=1}^m w_i + \sum_{i=1}^m u_i}$$

$H=1$: data is highly clustered

$H>0.75$: data is clustered at 90% confidence level

$H=0.5$: data is randomly distributed

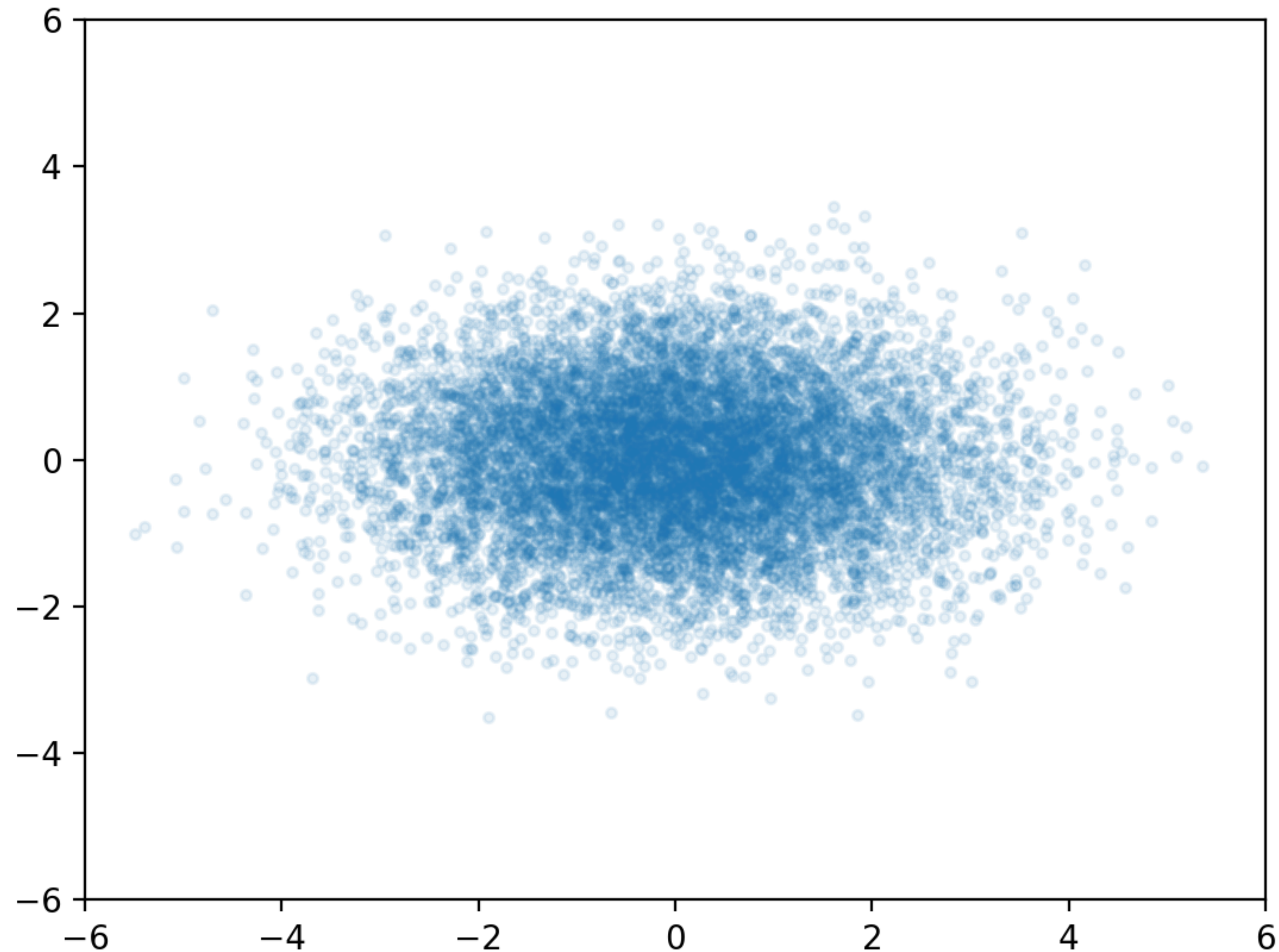
$H \approx 0$: data is uniformly distributed



The Hopkins statistic compares to a uniform distribution

Criticism of H:

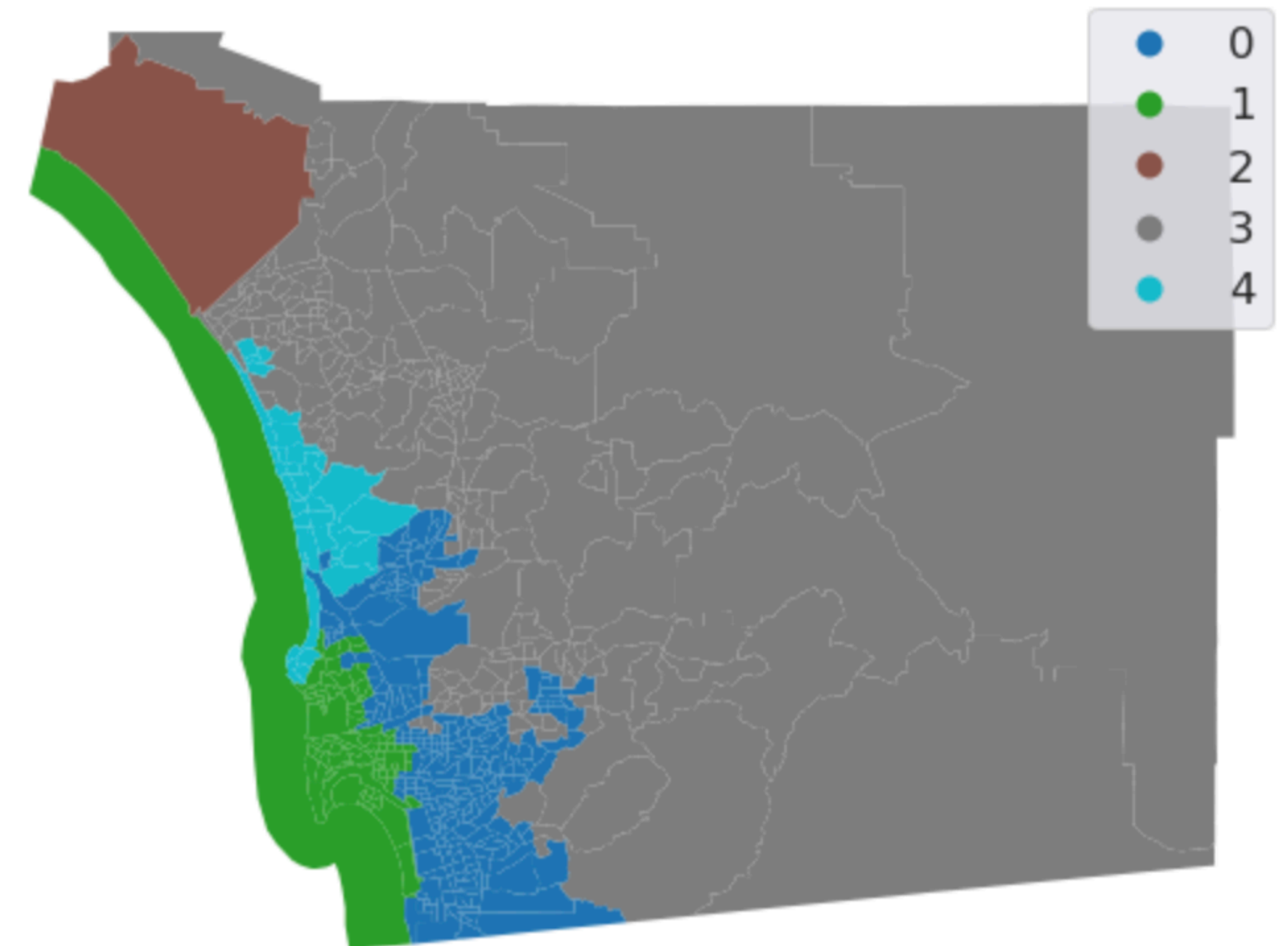
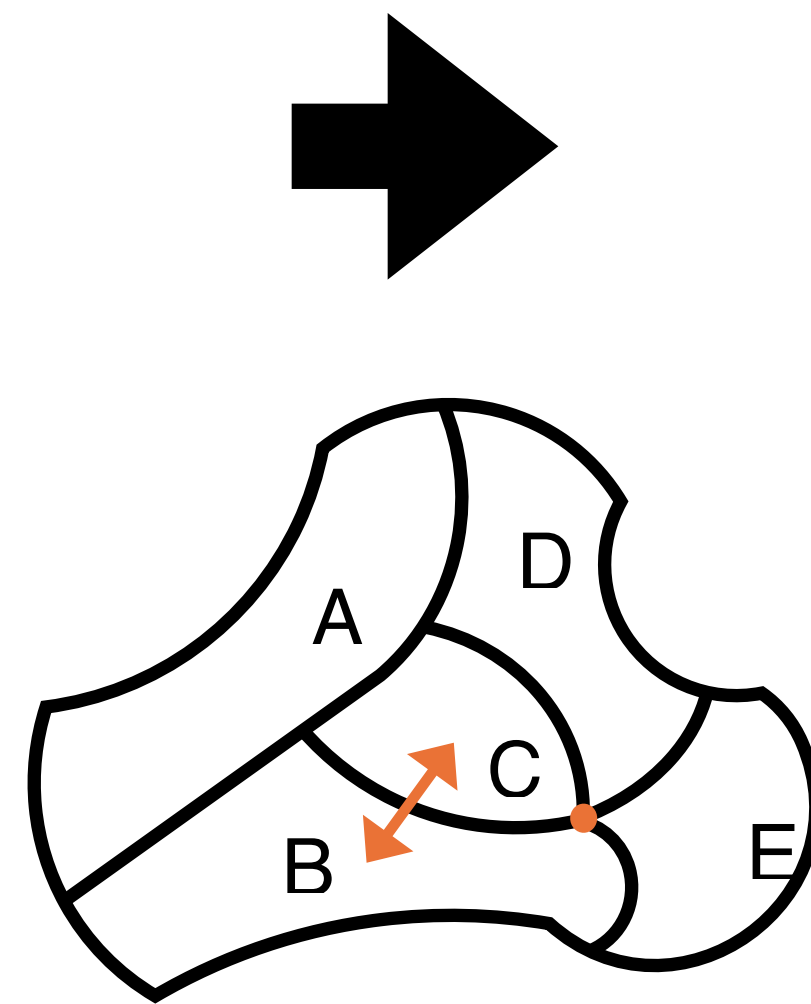
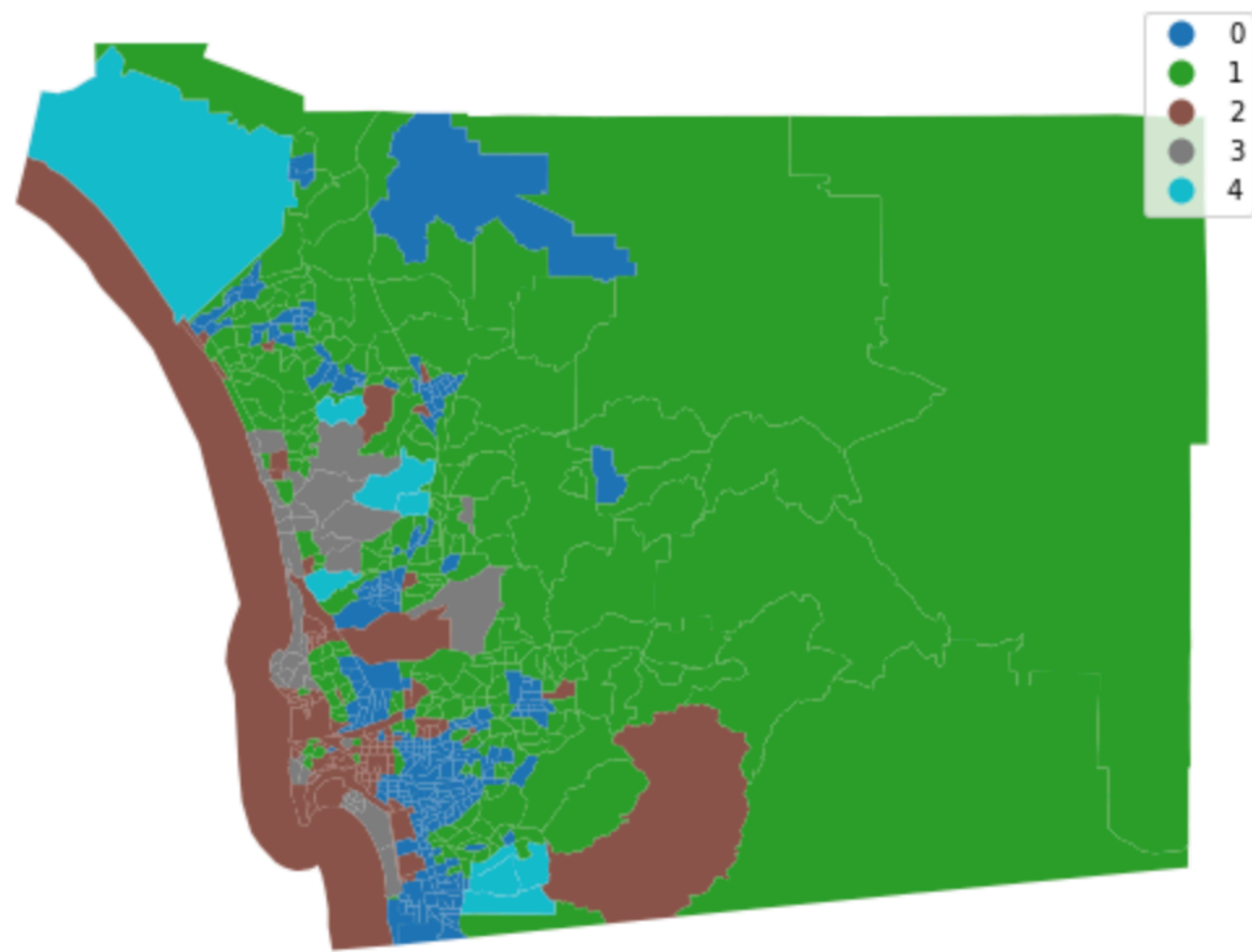
If there is just one "cluster", H is also close to 1



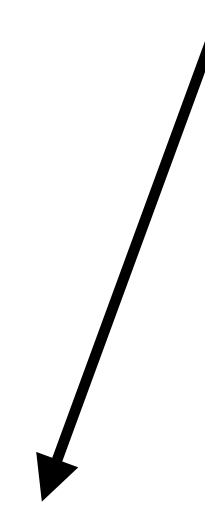
Spatially constrained clustering

Regionalization is "geographic clustering"

Regionalization is **spatially constrained clustering** where observations must be geographical neighbors to be able to be in the same cluster



Spatial contiguity constraint



You must be able to travel between all members in the cluster without traveling through non-member areas

Regionalization is "geographic clustering"

Regionalization works by **aggregation**

Predefined number of regions

Spatial **contiguity** constraint

Number of regions $<$ number of areas

No **overlapping** regions

No **empty** regions

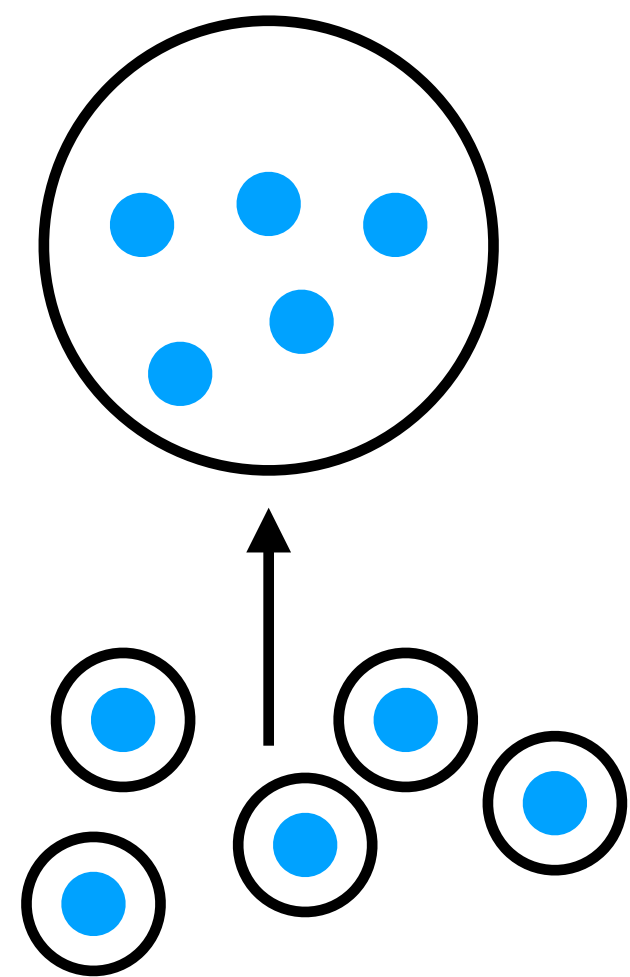
Regionalization is "geographic clustering"

Regionalization results in **analytical** or **functional** regions

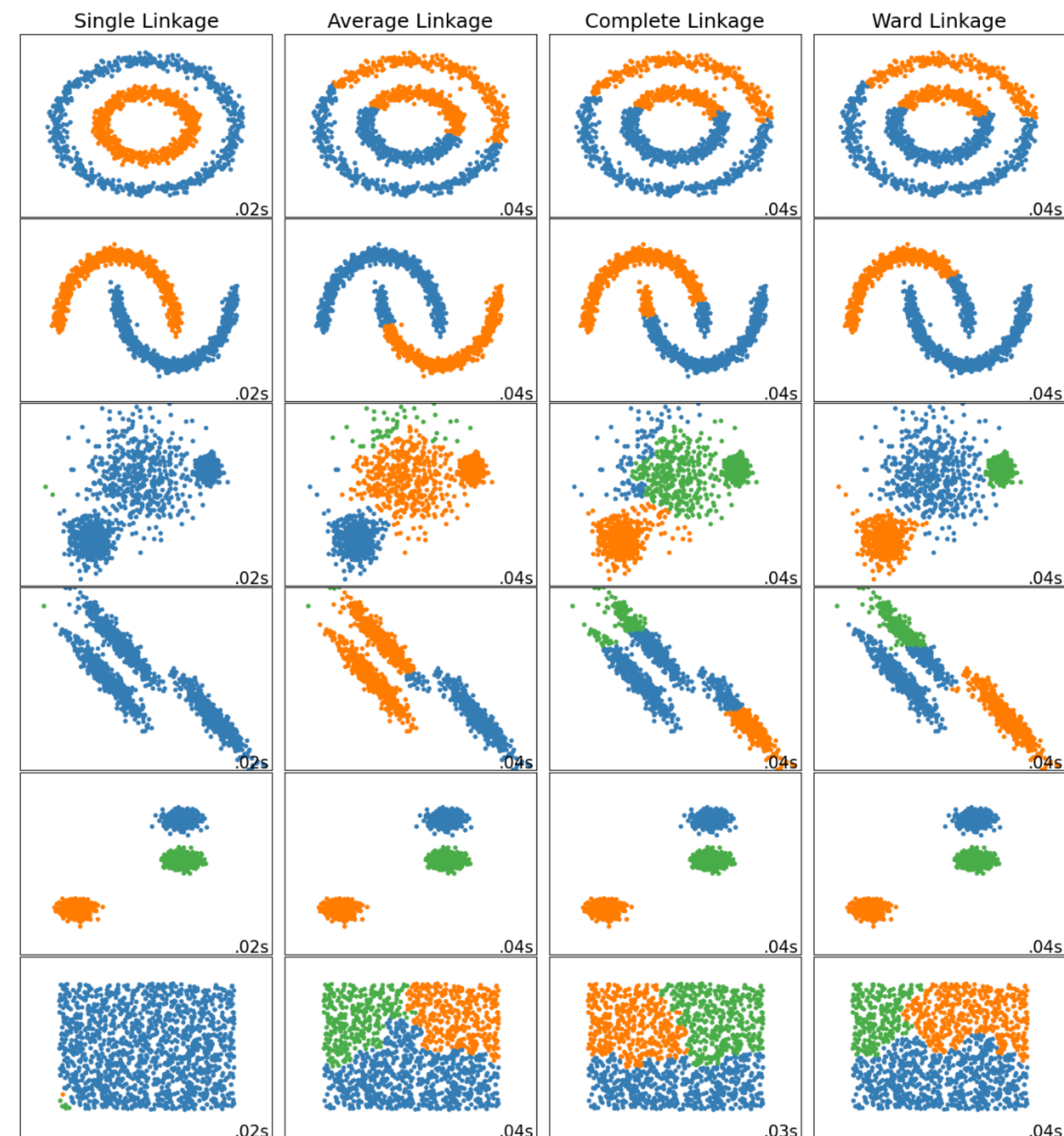
An alternative to normative regions and regular aggregation

Less prone to MAUP, ecological fallacy, etc.

Agglomerative hierarchical clustering (AHC)



...with **contiguity constraint**



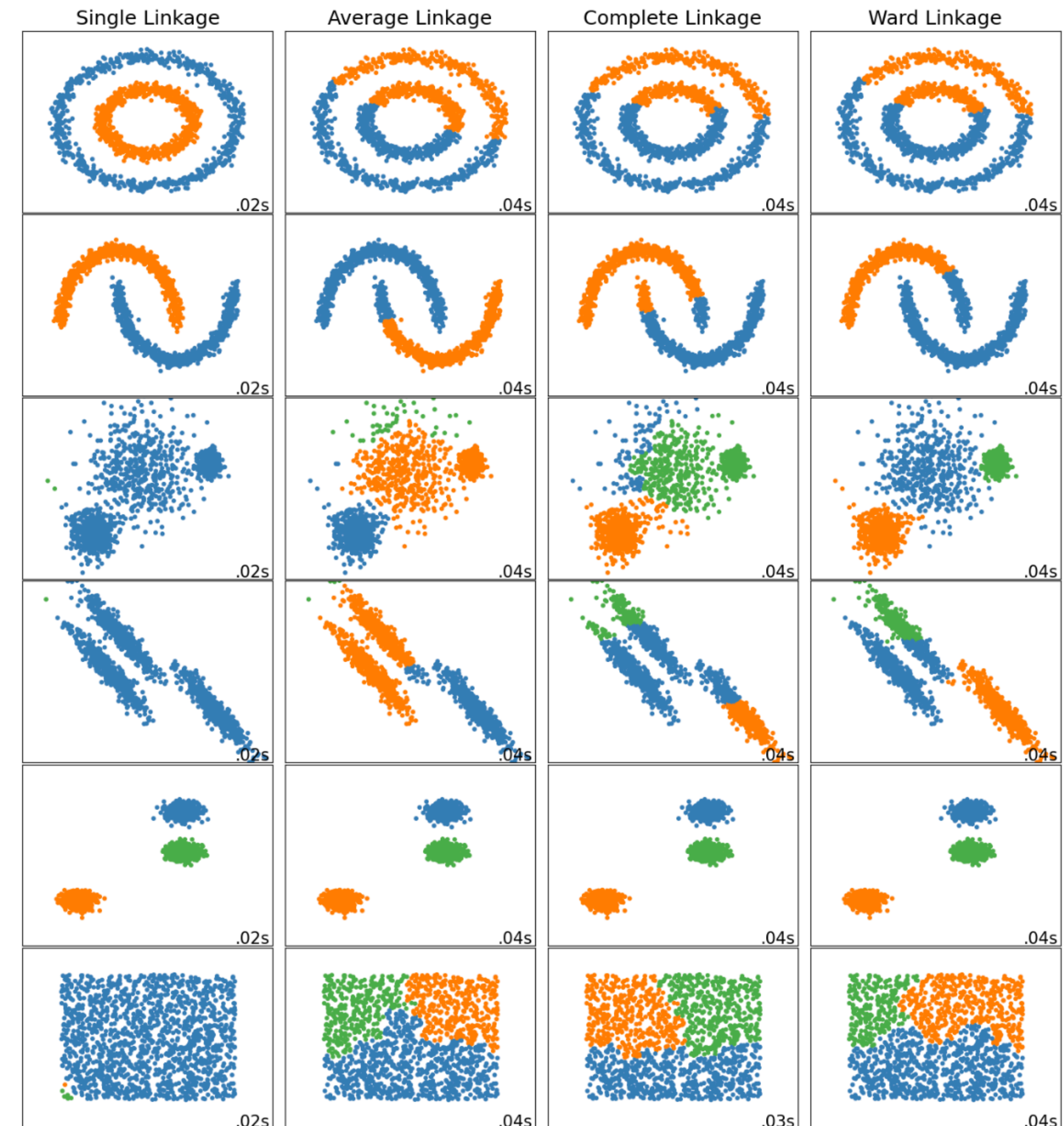
Agglomerative hierarchical clustering (AHC)

Ward minimizes the sum of squared differences within all clusters

Maximum or complete linkage minimizes the maximum distance between observations of pairs of clusters

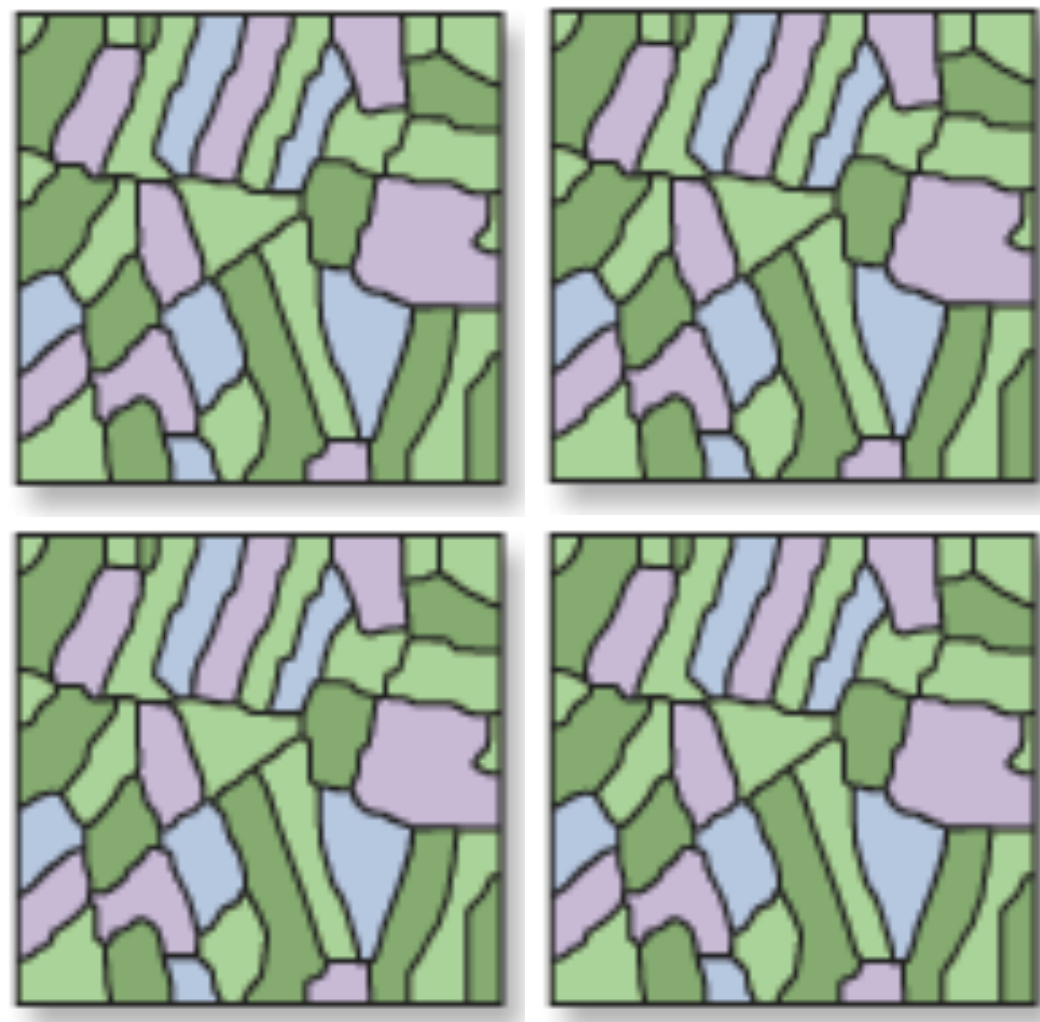
Average linkage minimizes the average of the distances between all observations of pairs of cluster

Single linkage minimizes the distance between the closest observations of pairs of clusters



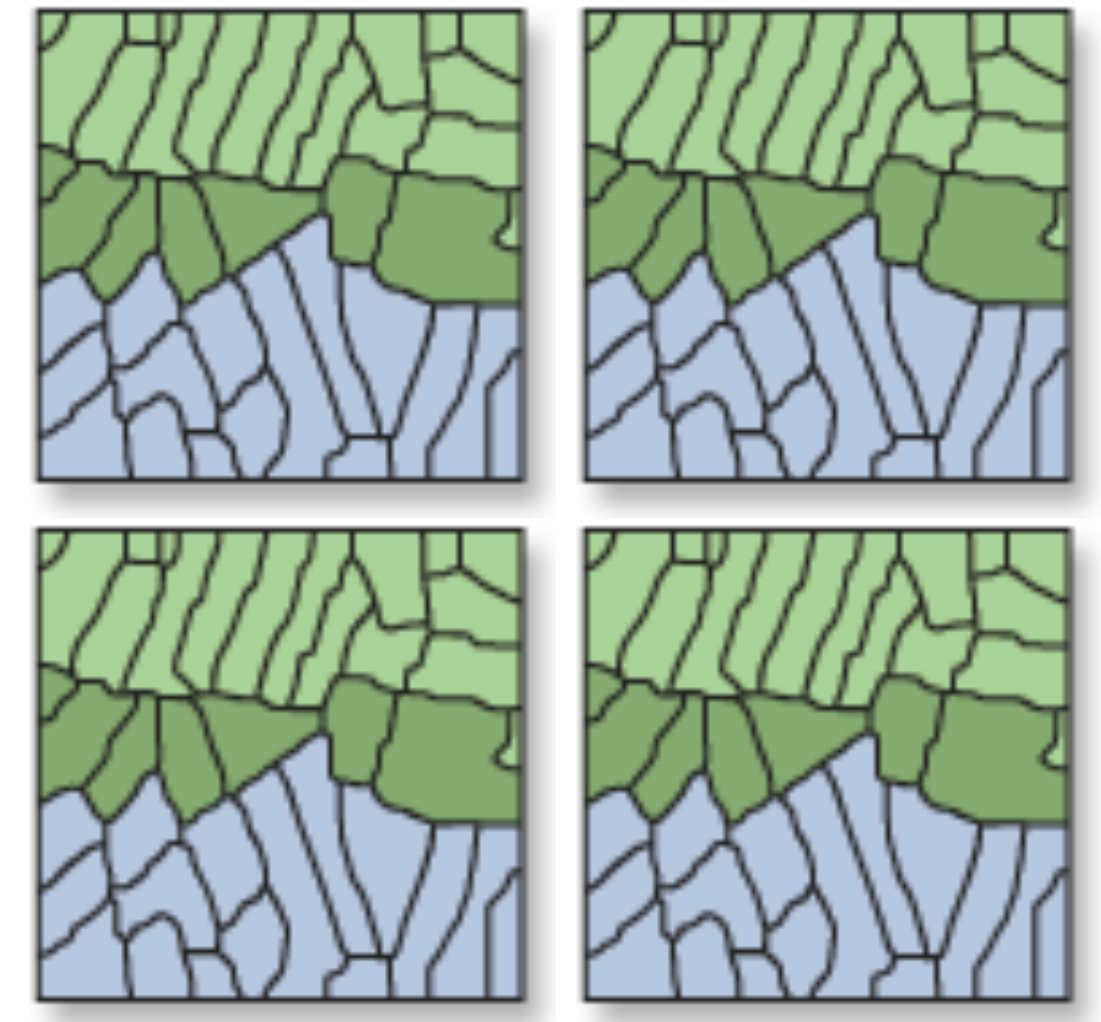
Check individual autocorrelations before spatial clustering

Low



Gives bad fit,
too small regions

High

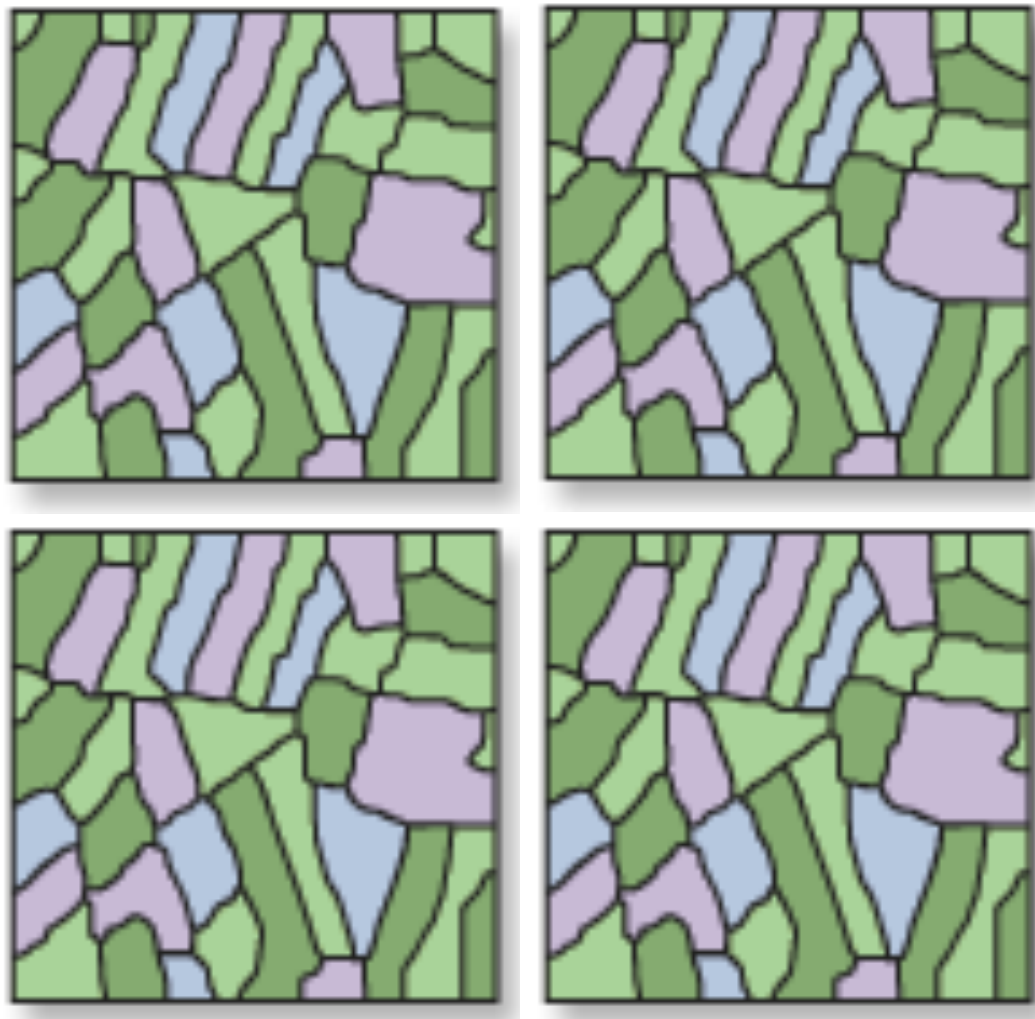


Non-spatial clustering
might be good enough



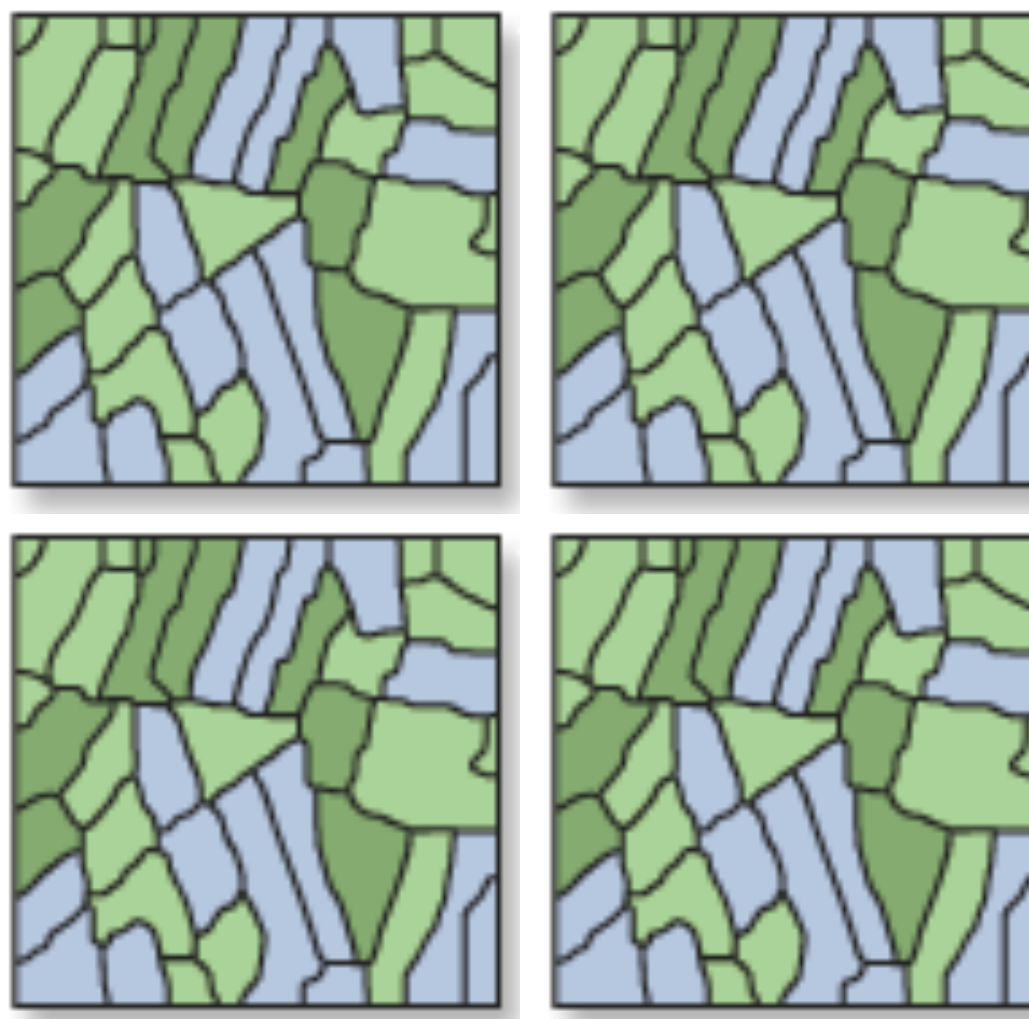
Check individual autocorrelations before spatial clustering

Low



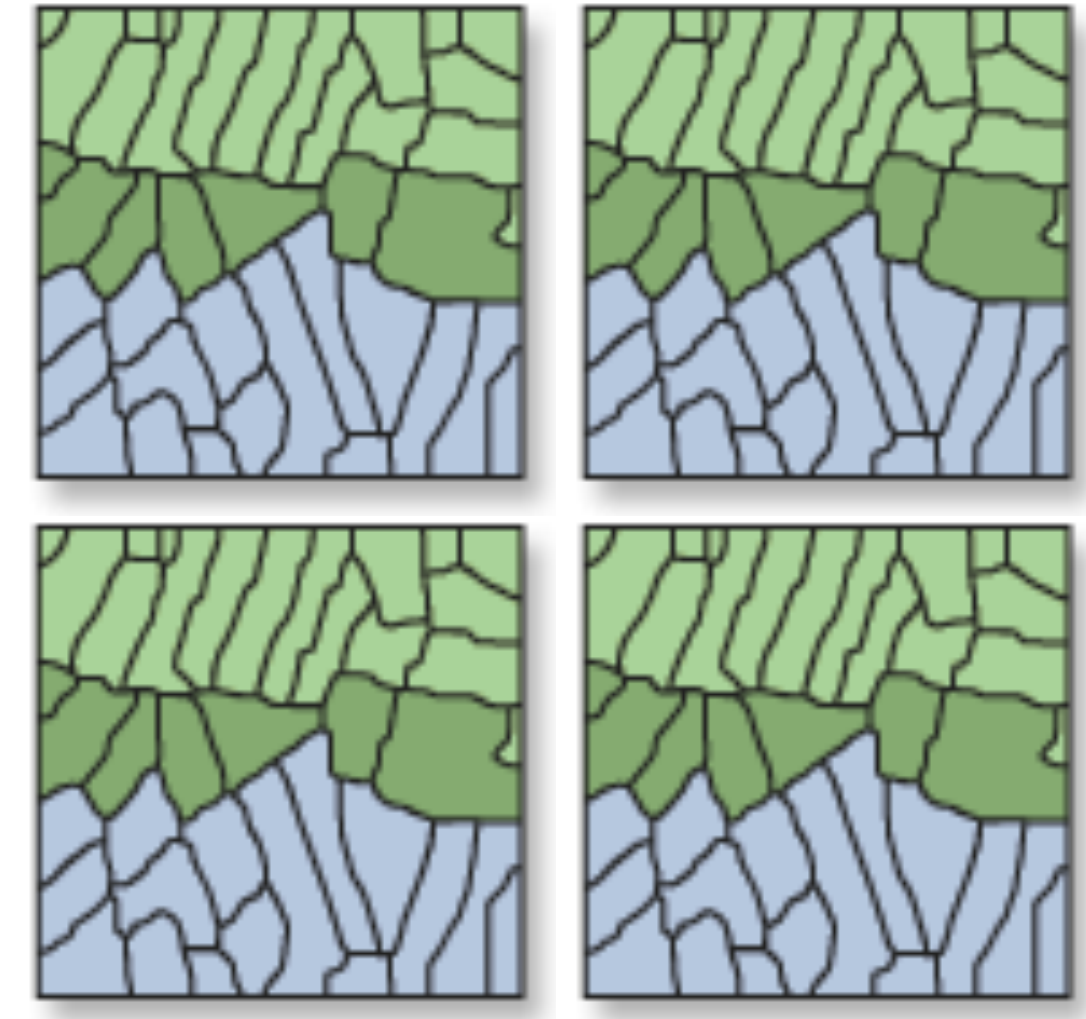
Gives bad fit,
too small regions

Sweet spot



Spatial clustering
is meaningful

High



Non-spatial clustering
might be good enough

In any case, spatial constraints usually lower 'goodness of fit'

Comparing goodness of fit / feature coherence

Calinski-Harabasz Index

“the within-cluster variance divided by the between-cluster variance”

Silhouette Score

“the average standardized distance from each observation to its “next best fit” cluster”

and more...

Metrics for solution similarity

To compare **normative** with **data-driven** regions



Metrics for solution similarity

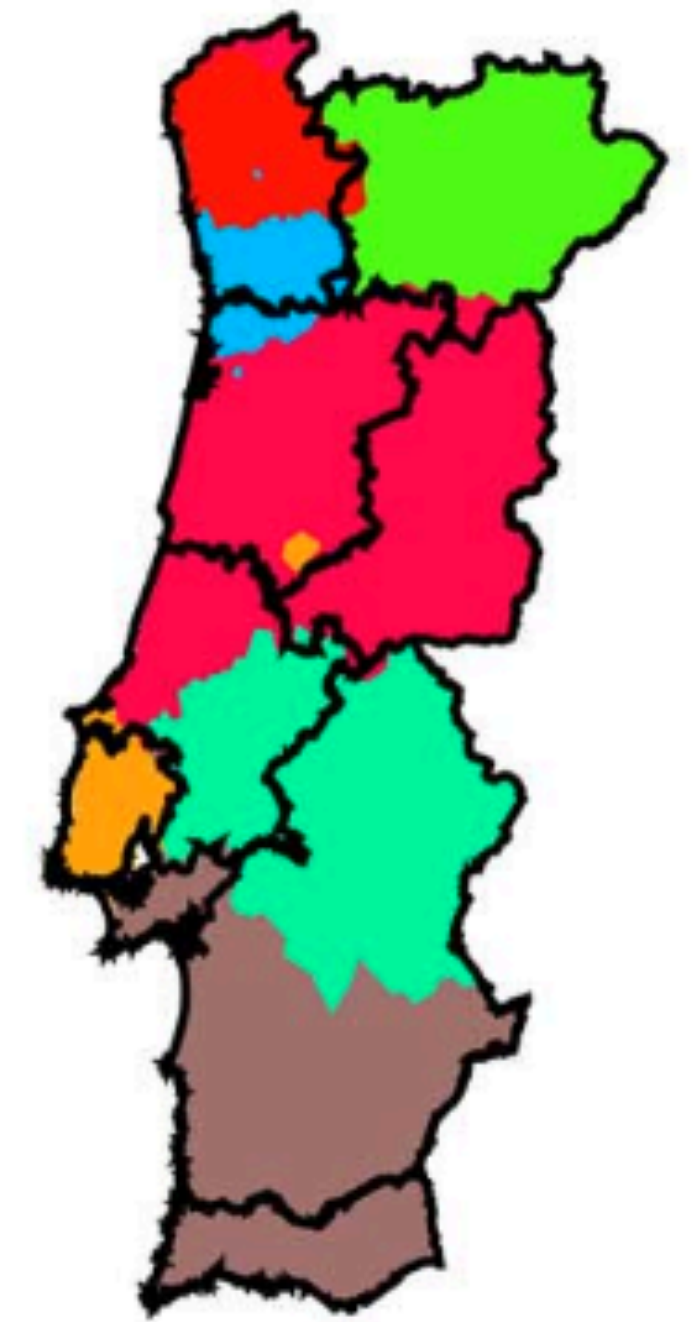
To compare **normative** with **data-driven** regions



Today



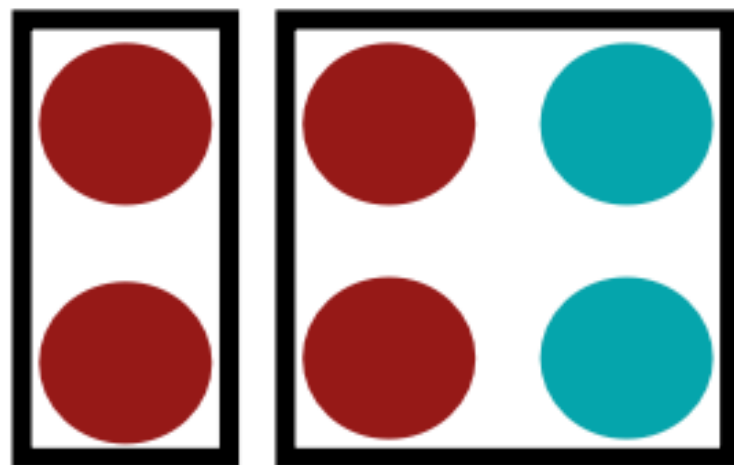
Historic



Failed
referendum

Metrics for solution similarity between partitions

- n_{11} number of pairs of elements in the same community under both X and Y
- n_{00} number of pairs of elements not in the same community under both X and Y
- n_{01} number of pairs of elements not in the same community under X but in the same community under Y
- n_{10} number of pairs of elements in the same community under X but not in the same community under Y

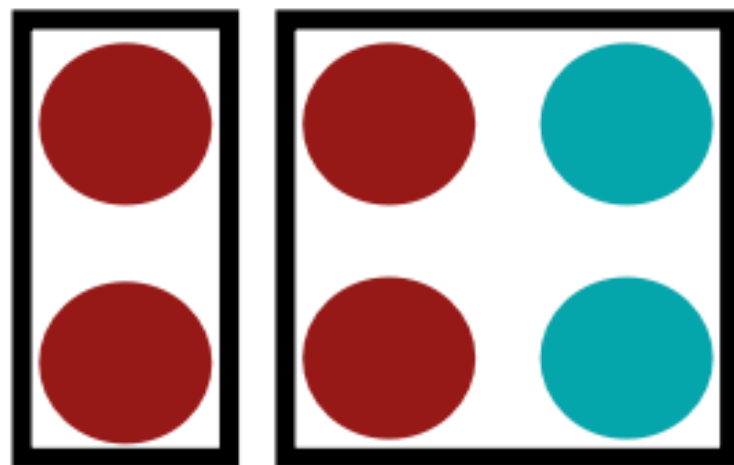


X **data-driven**

Y **normative**

Metrics for solution similarity between partitions

3	n_{11}	number of pairs of elements in the same community under both X and Y
4	n_{00}	number of pairs of elements not in the same community under both X and Y
4	n_{01}	number of pairs of elements not in the same community under X but in the same community under Y
4	n_{10}	number of pairs of elements in the same community under X but not in the same community under Y



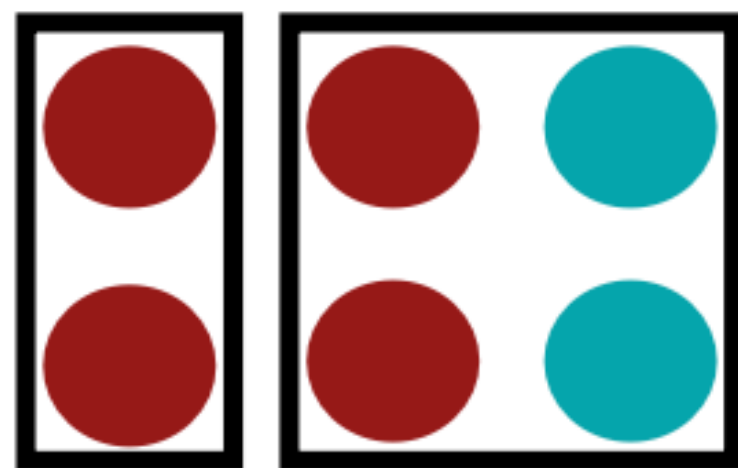
X **data-driven**

Y **normative**

Metrics for solution similarity between partitions

Rand Index

3	n_{11}	number of pairs of elements in the same community under both X and Y
4	n_{00}	number of pairs of elements not in the same community under both X and Y
4	n_{01}	number of pairs of elements not in the same community under X but in the same community under Y
4	n_{10}	number of pairs of elements in the same community under X but not in the same community under Y



X **data-driven**

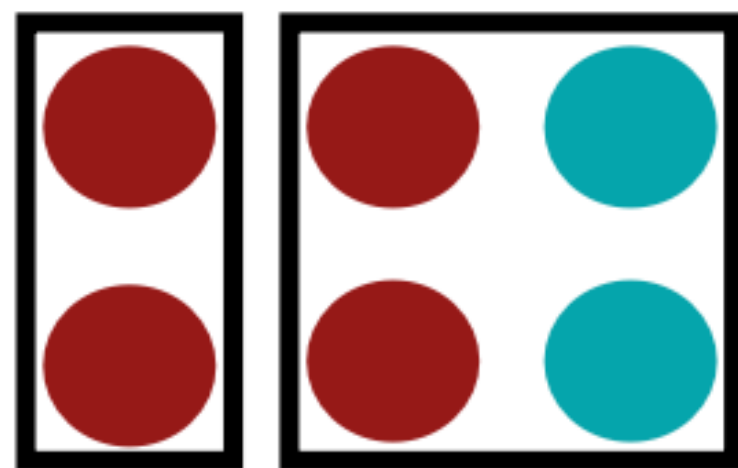
Y **normative**

$$\mathcal{R} = \frac{n_{00} + n_{11}}{n_{00} + n_{11} + n_{01} + n_{10}} = 0.47$$

Metrics for solution similarity between partitions

Fowlkes-Mallows Index

3	n_{11}	number of pairs of elements in the same community under both X and Y
4	n_{00}	number of pairs of elements not in the same community under both X and Y
4	n_{01}	number of pairs of elements not in the same community under X but in the same community under Y
4	n_{10}	number of pairs of elements in the same community under X but not in the same community under Y



X **data-driven**

Y **normative**

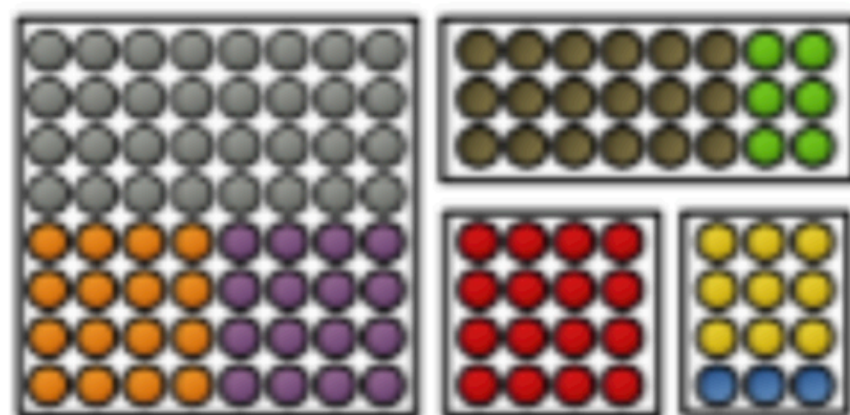
$$\mathcal{F} = \sqrt{\frac{n_{11}}{n_{11} + n_{10}} \cdot \frac{n_{11}}{n_{11} + n_{01}}} = 0.43$$

Metrics for solution similarity between partitions

Adjusted Rand Index

$$ARI = \frac{n_{00} + n_{11} - \mathbf{E}[n_{00} + n_{11}]}{n_{00} + n_{11} + n_{01} + n_{10} - \mathbf{E}[n_{00} + n_{11}]}$$

where $\mathbf{E}[n_{00} + n_{11}]$ are the values expected by chance.
This index can be negative.



$$ARI = 1.00$$

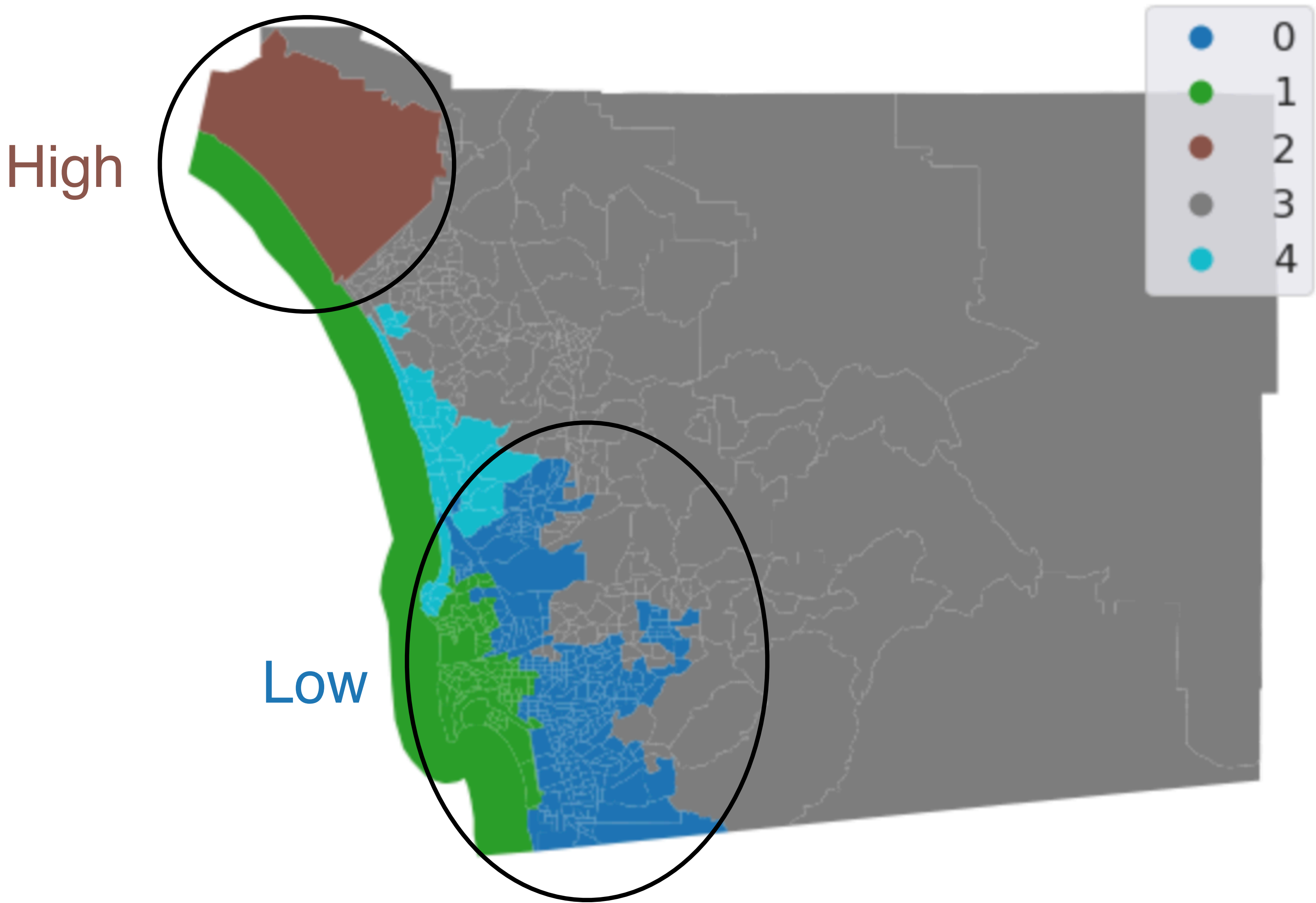


$$ARI = 0.88$$



$$ARI = 0.03$$

Geographical coherence measures spatial performance



Geographical coherence measures spatial performance

Blackboard

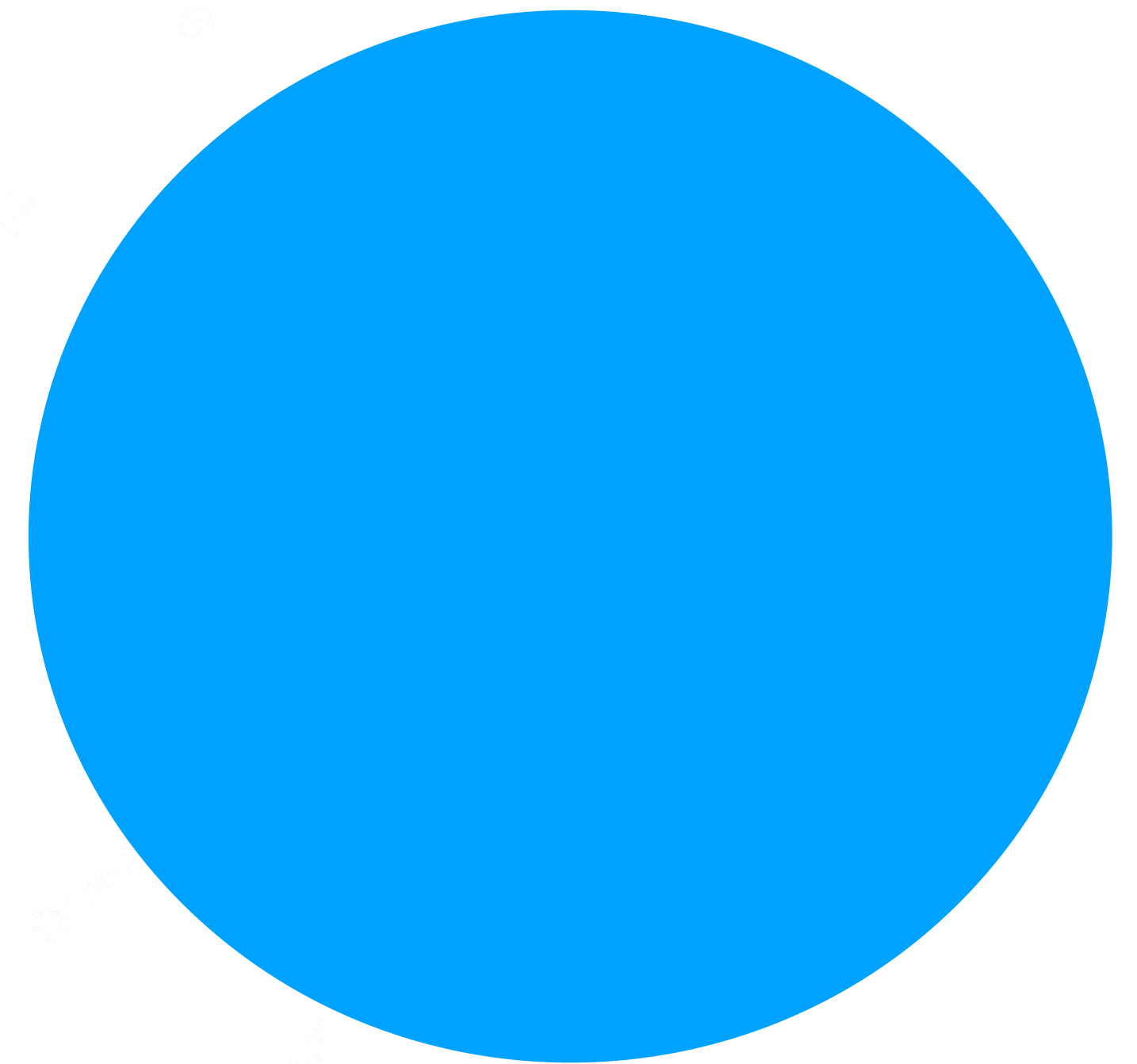
Isoperimetric quotient compares the area of the region to the area of a circle with the same perimeter as the region

Area of region

$$IPQ = \frac{A}{A_c} = \frac{4\pi A}{P^2}$$

Area of circle with same perimeter

Perimeter of region



Isoperimetric quotient



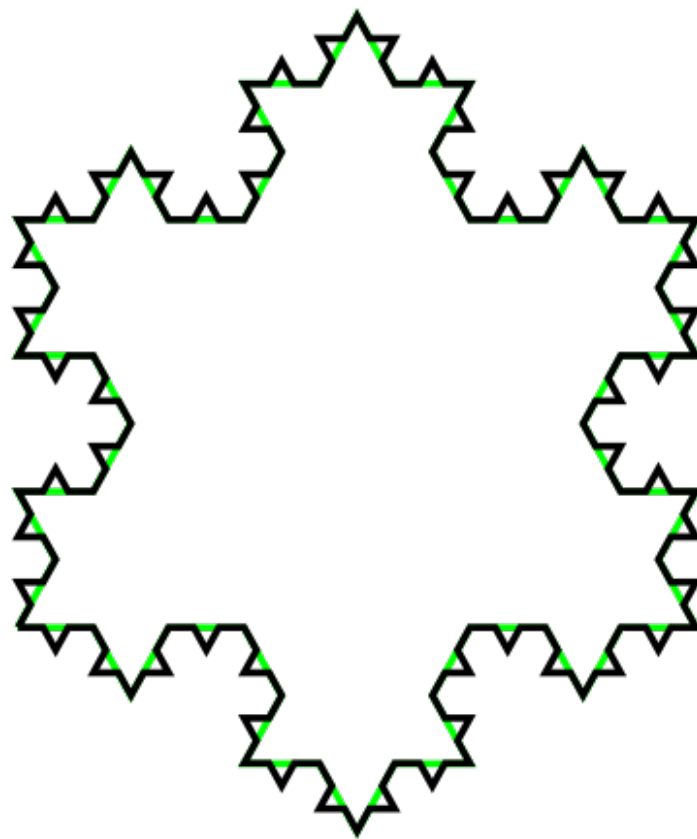
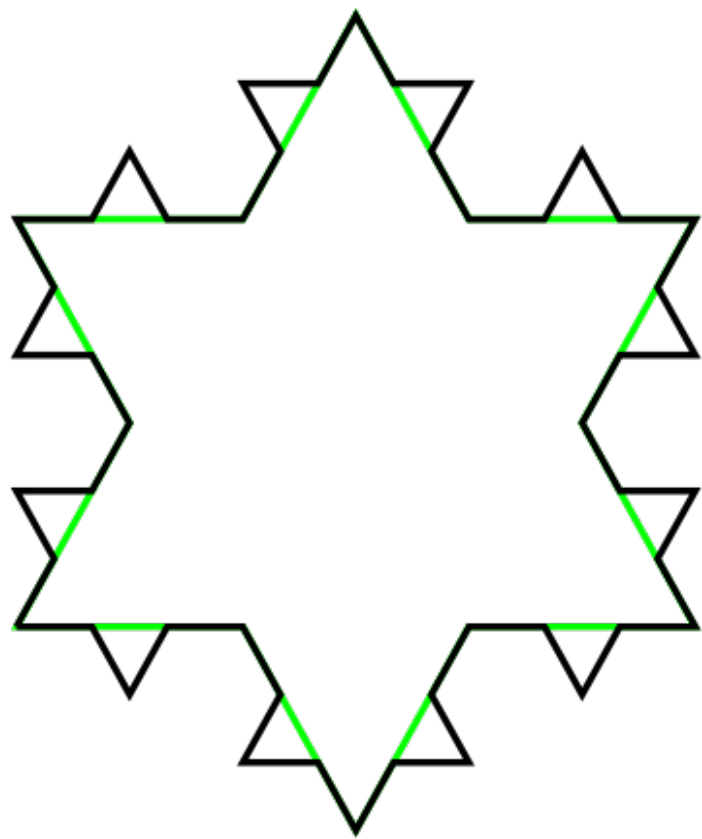
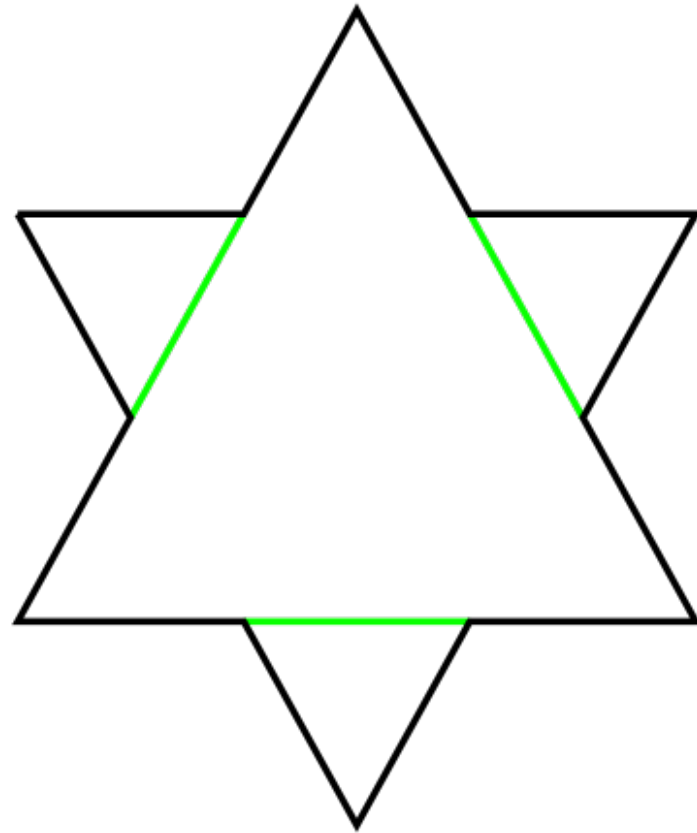
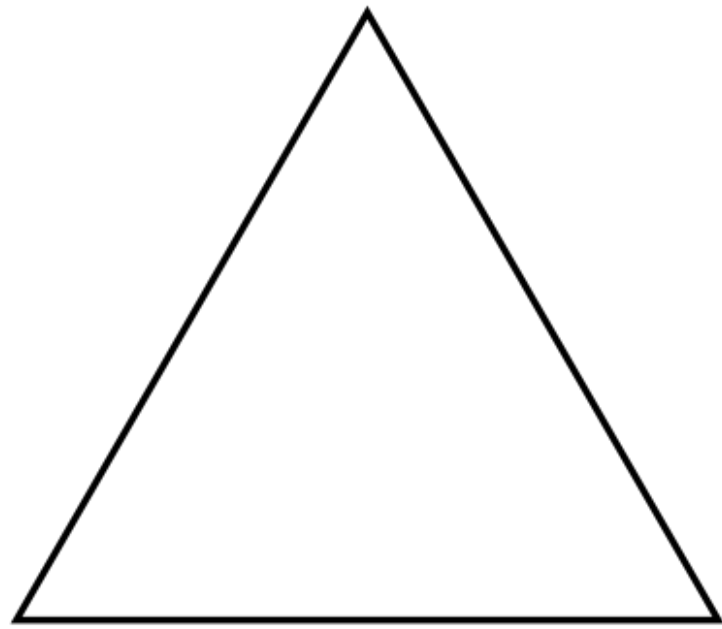
IPQ closer to 0



IPQ closer to 1

Sidenote: Perimeter might not be a great measure

Blackboard

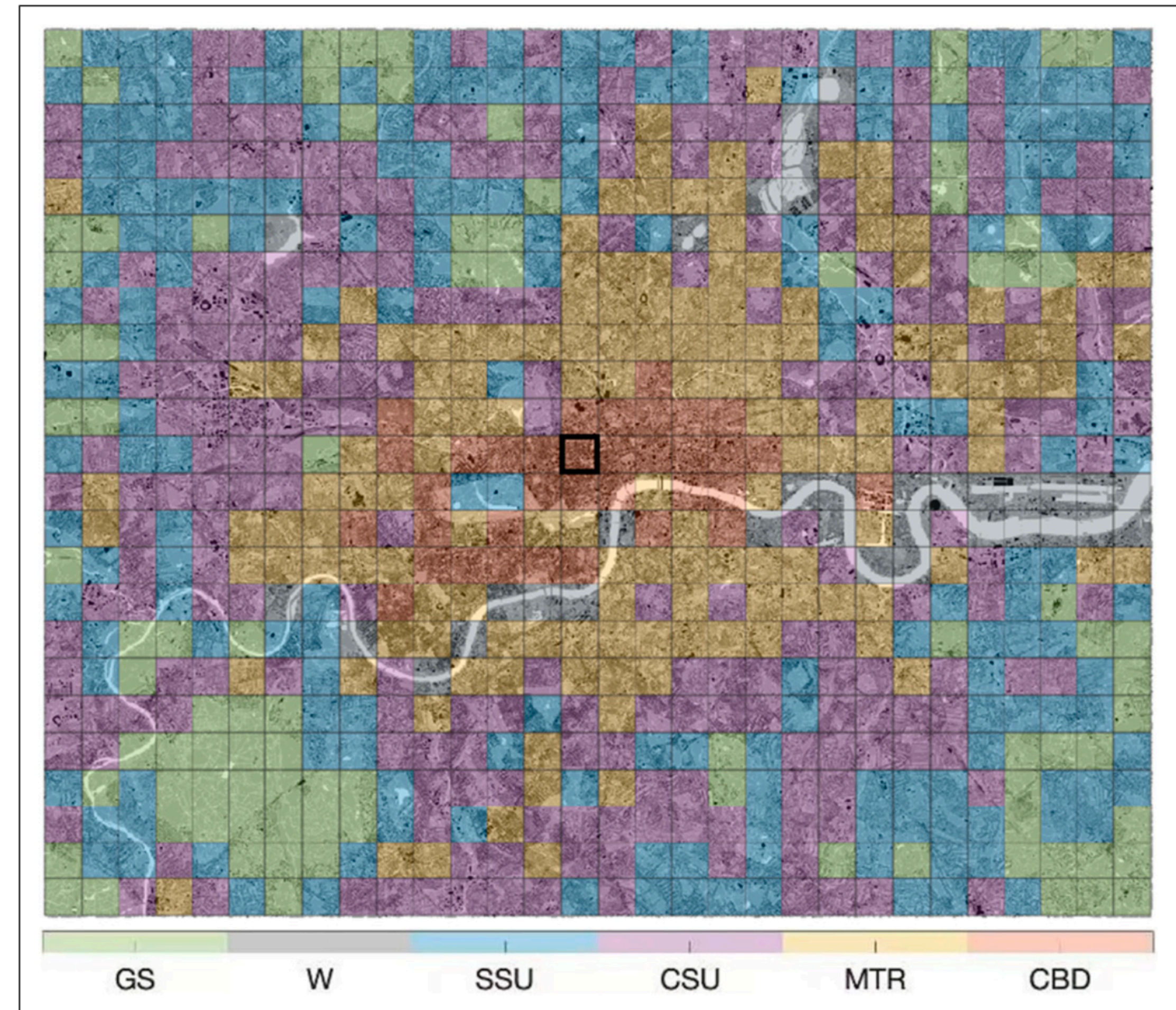
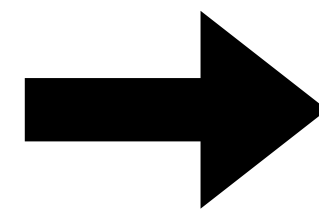
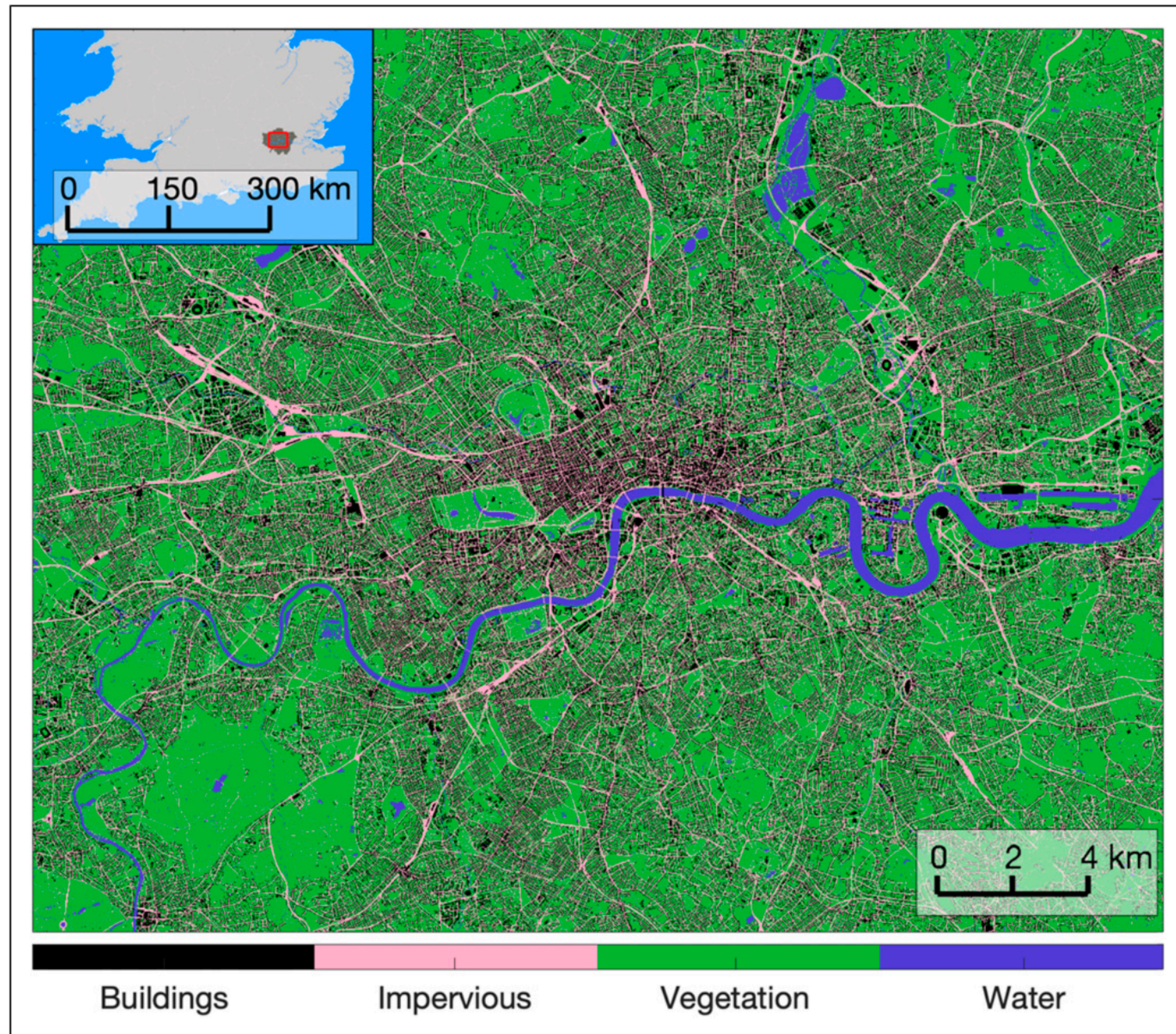


Sidenote: Perimeter might not be a great measure



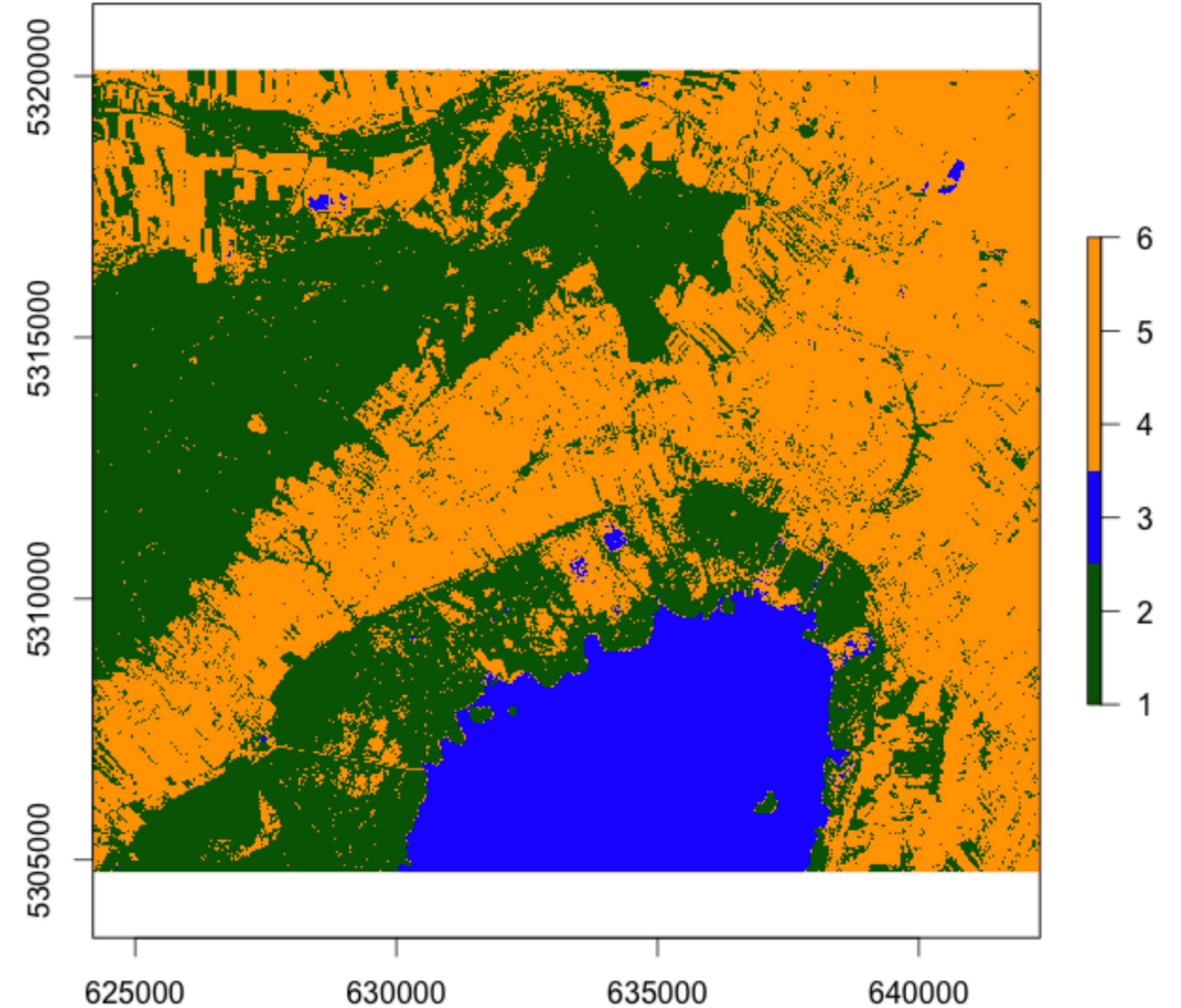
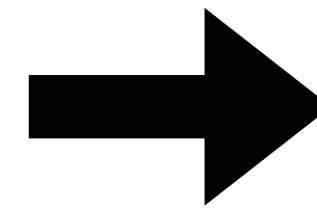
When do we use clustering in GDS?

Identify similar urban areas



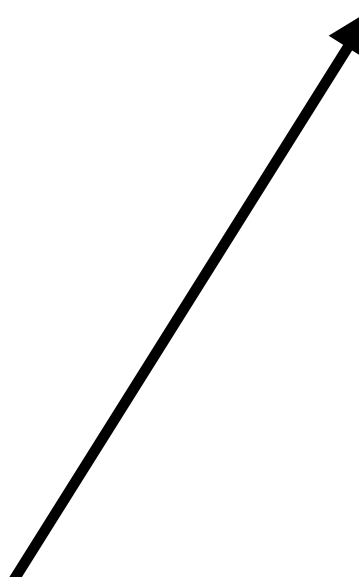
When do we use clustering in GDS?

Processing satellite images




```
sagg12 = cluster.AgglomerativeClustering(n_clusters=12, connectivity=w.sparse)
```

spatial constraint

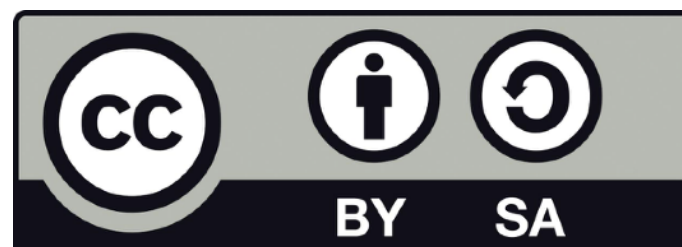


Jupyter

Sources and further materials for today's class



***Geographic Data Science
with Python***



[https://geographicdata.science/book/notebooks/
10_clustering_and_regionalization.html](https://geographicdata.science/book/notebooks/10_clustering_and_regionalization.html)

https://darribas.org/gds_course/content/bG/concepts_G.html

Next week: Point Pattern Analysis

